Evidence Supplement for From Delegation to Audit: A Systematic Review of Cryptographic Assurance in LLM Agents

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Detailed evidence tables, RQ visualizations, search expressions, extraction-batch accounting, and recorded claim appraisals supporting the compact main article. This document accompanies canonical release 3.6.3 and the explicitly post hoc CS1 analyses, with a separate R3 ledger of targeted source clarifications and additional descriptive sensitivities. It does not report a new whole-corpus adjudication or replication of primary experiments. Scientific claims and the JAIR reproducibility checklist are presented in the main article.

1 Scope and reading guide

This supplement preserves the detailed numerical tables and source-specific qualifications underlying the main article (revision R3, 4 November 2026). The numerical baseline remains canonical release 3.6.3 and the CS1 analyses. R3 adds targeted source checks and descriptive sensitivities in a separate ledger; it does not change canonical extraction labels or reproduce primary experiments.

Tables S1–S22 retain the analytical tables; Tables S23–S27 accompany the search/batch/appraisal appendices. Figures S1–S4 provide the RQ visualizations. The machine-readable package supplies data, locators, dictionaries, normalized screening decisions and scripts. The main article contains the substantive arguments and limitations needed to assess the review without these inventories.

Source clarifications and methodological status in R3

The source-check ledger, `results/source_checks_R3.md`, distinguishes confirmation of published passages from reproduction of an artifact. It also qualifies wording retained in the historical extraction. In PCC-LLM's Pilot C, 12/18 is the blocking fraction over adversarial proposals: five allowed probes remain within the specified budget and one concerns an omitted predicate. The failed preregistered target is retained; the six allowed proposals are not six violations of the formal checker specification (Koomullil 2026). A2ABreak identifies Alloy and TLA+/TLC and explicitly excludes ProVerif; Table S25 now uses that source attribution (Lotfi et al. 2026).

R3's post hoc script, `scripts/jair_revision_analysis.py`, gives equal weight to reports when summarizing within-role support: 74.0% across 328 mechanism reports and 95.4% across 218 attack/benchmark reports. It also excludes three timer-discordant READY observations from the latency sensitivity, leaving 359. The L3 median changes from 262.88 to 920.95 ms (40 to 38 rows), and L4 from 1,700 to 1,641.4 ms (193 to 192 rows); L1/L2 are unchanged. Tables S12–S13 retain the original baseline. Both readings are documented in `results/jair_latency_sensitivity.csv`; no row is silently reassigned.

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The authors confirmed independent human source assessment before adjudication. The subsequently supplied Stage 3 archive contains 533 R1 and 508 R2 rows over 442 reports. The first-row comparison and the supplied script were reproduced, with independent contingency tables and pair locators. Grouped PQC is 431/442 (97.51%; $\kappa = .7320$, AC1=.9726), correcting the earlier 427/442 summary. The support coefficients ($\kappa = .2878$, AC1=.8074; 367/442 matches) concern three recorded labels, including NOT_EVALUATED, rather than a binary outcome. The first-row comparison is not agreement over every extracted claim.

Exact boundary-set agreement is 149/442 (33.71%), and UG agreement is 180/442 (40.72%; $\kappa = .0934$, AC1=−.1831). These low values persist in the 319 single-claim-report subset and 474 ordered pairs. The latter leave 59 R1 and 34 R2 claims unmatched. Maturity has eight observed labels including NE; the supplied aggregation rank table recognizes none of the actual maturity labels, so its "highest maturity" output is not used. Raw-pair results, missing-state sensitivities and script qualifications are documented in `results/stage3_irr_reproduction_report.md`. The raw archive remains unchanged under `stage3_irr/`.

R1 metadata retain 148 UNASSIGNED labels and 212 AI-draft rationales. The authors confirm that Herman Kabetta checked and approved these worksheets before adjudication; the approval notes are still being located. Approval is author-confirmed, whereas the worksheet calculations have been reproduced. The initial summary and binary-feasibility audit remain as superseded history. The search strategy accounts for 4,093 relevance removals as 728 ACM plus 3,365 arXiv entries, separately from 1,182 duplicates. The authors confirm no subsequent audit of those removals. Reconciled accounting does not establish correct exclusion decisions or estimate filter recall.

2 Detailed analytical tables

Table S1. Four analytical assurance boundaries and their observable evidence targets.

Boundary	Operational transition	Assurance question	Relevant observable or evidence target
B1	User intent → delegation/policy	Is the authority expressed in the delegated policy appropriately constrained by the user's represented task?	Task representation, policy translation, consent or approval artefact, scope of delegated authority
B2	Delegation/policy → agent request	Is a request bound to an authentic and currently applicable delegation, scope, and policy state?	Authorization evidence, subject/resource/action binding, caveats, expiry, revocation and replay controls
B3	Authorized request → actual tool/runtime execution	Does the enforced authorization correspond to the operation and parameters actually executed?	Enforcement placement, request/operation binding, trusted execution measurements or traces, state transitions
B4	Execution → verifiable audit evidence	What aspects of the execution history can an independent verifier establish from recorded evidence?	Event capture, provenance, origin authenticity, record integrity, ordering, coverage and completeness conditions

Table S2. Scope of related reviews and the emphasis of this review.

Related perspective	Question it informs here	Additional object of this review
Threat and protocol taxonomies (Deng et al. 2025; Gan et al. 2026; Tamayo 2026; Umer et al. 2026; M. Yu et al. 2025)	Which attack surfaces and defenses recur?	Whether a particular claim denotes an attack finding, a mechanism guarantee, or an empirical measurement
Trust–authorization mismatch (G. Shi et al. 2026)	How does interpreted intent differ from permitted action?	The protected representation, enforcement point, and assumptions supporting each B1/B2 claim
Operational auditability (Theodorakopoulos and Theodoropoulou 2026)	What execution evidence supports accountability?	The distinction between integrity of captured records and completeness of event capture at B4
Confidential computing (Forough et al. 2026)	What can an attested execution environment protect?	The correspondence between attested code, tested implementation, policy meaning, and attestation dependencies
AI and cryptography (Bakshi et al. 2026)	Which cryptographic functions are relevant?	Claim-linked constructions, migration evidence, and missing protection-lifetime information

Table S3. Mutually exclusive primary reasons for full-text exclusion.

Primary full-text exclusion code	Meaning	Reports
`EC1_NON_AGENTIC`	Outside the defined agentic-system scope	53
`EC2_OFFENSIVE_ONLY`	Offensive use without the required agent-system assurance contribution	3
`EC4_INCIDENTAL_APP`	Incidental domain application without the required contribution	76
`EC5_SECONDARY_OR_CONTAINER`	Secondary-only or ineligible container contribution	19
`LANGUAGE_NOT_ENGLISH`	English full-text requirement not met	1
`EC_RIGOR_1`	Preprint rigor exclusion, subtype 1 as recorded	9
`EC_RIGOR_2`	Preprint rigor exclusion, subtype 2 as recorded	5
Total	**152 substantive/language exclusions + 14 rigor exclusions**	**166**

Table S4. Operational appraisal categories and their evidentiary interpretation.

Code	Assign when	Interpretation limit
DG	Direct empirical or analytical evidence demonstrates failure of an identified property on an identified target	Does not establish a primitive break or failure of an untested defense
AD	A material trust/TCB, deployment or model–implementation dependency bounds the assessed claim	Ordinary mathematical assumptions do not automatically constitute a discovered gap

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Table S4. Operational appraisal categories and their evidentiary interpretation. (continued)

Code	Assign when	Interpretation limit
UG	An identifiable author claim extends beyond the support offered by its model, proof or experiments	An extractor's error, explicit non-goal or unattempted unrelated property is not author overclaim
NG	Completed assessment identifies no DG, AD or UG within the explicit claim and scope	Exclusive; neither absence of all vulnerabilities nor a universal requirement for both proof and prototype
NE / NA	Assessment unfinished / rubric inapplicable	Neither is a positive assurance or vulnerability finding

Table S5. Included reports and claims by evidence tier, stream, and publication year.

Corpus dimension	Tier 1 studies (E01–E04)	Tier 2 studies (E05–E09)	Total studies (n = 442)	Study share (%)	Tier 1 claims (n = 271)	Tier 2 claims (n = 313)	Total claims (n = 584)	Claim share (%)	Claims per study
All included studies	**212**	**230**	**442**	**100.0%**	**271**	**313**	**584**	**100.0%**	**1.32**
Stream A (*Agentic AI Security & Boundary Attacks*)	99	129	**228**	**51.6%**	131	159	**290**	**49.7%**	1.27
Stream M (*Cryptographic / Hardware Mechanisms & MCP*)	68	39	**107**	**24.2%**	78	51	**129**	**22.1%**	1.21
Stream M_AND_A (*Integrated Mechanism & Attack Evaluation*)	45	62	**107**	**24.2%**	62	103	**165**	**28.3%**	1.54
Publication Year: 2023–2024 (*Early Tool-Using Agents*)	3	0	**3**	**0.7%**	5	0	**5**	**0.9%**	1.67
Publication Year: 2025 (*First Wave of MCP & A2A Protocols*)	29	38	**67**	**15.2%**	33	46	**79**	**13.5%**	1.18
Publication Year: 2026 (+2 Early-Access 2027) (*Ecosystem Expansion*)	180	192	**372**	**84.2%**	233	267	**500**	**85.6%**	1.34

Table S6. Claim categories and boundary cardinality, for all and READY claims.

Claim category (Claim_Category)	MULTI_BOUNDARY (≥ 2 boundaries in Boundary_Codes)	Multi-boundary share (%)	B1 single (Intent → Delegation)	B2 single (Delegation → Request)	B3 single (Request → Execution)	B4 single (Execution → Audit)	Total claims (n = 584)	READY subset (n = 530)
DEFENSIVE_MECHANISM	**235**	85.5%	1	11	16	12	**275 (47.1%)**	241 (45.5%)
EMPIRICAL_BENCHMARK	**81**	64.3%	0	6	31	8	**126 (21.6%)**	121 (22.8%)
ATTACK_FINDING	**79**	88.8%	0	5	5	0	**89 (15.2%)**	86 (16.2%)
FORMAL_FRAMEWORK	**34**	91.9%	0	1	0	2	**37 (6.3%)**	32 (6.0%)
SYSTEM_PROTOTYPE	**25**	73.5%	0	2	2	5	**34 (5.8%)**	29 (5.5%)
VULNERABILITY_ATTACK_SURFACE	**18**	78.3%	0	0	1	4	**23 (3.9%)**	21 (4.0%)
Total across all 584 claims	**472**	**80.8%**	**1 (0.2%)**	**25 (4.3%)**	**55 (9.4%)**	**31 (5.3%)**	**584 (100.0%)**	—
QA-Released READY subset (n = 530)	**425**	**80.2%**	**1 (0.2%)**	**24 (4.5%)**	**52 (9.8%)**	**28 (5.3%)**	—	**530 (100.0%)**

Table S7. Individual boundary incidence and exact boundary combinations.

Analysis dimension	Boundary code or exact combination	Claim count (n = 584)	Claim incidence (%)	Unique study count (n = 442)	Study incidence (%)	Dominant architectural role in the corpus
Individual boundary incidence	**B3 (B3_REQUEST_TO_EXECUTION)**	**491**	**84.1%**	**395**	**89.4%**	Pre-execution tool-call gating, sandbox isolation (WASM/eBPF/TEE), and parameter binding
Individual boundary incidence	**B2 (B2_DELEGATION_TO_REQUEST)**	**472**	**80.8%**	**398**	**90.0%**	Capability token propagation, context/memory taint tracking, and MCP/A2A request mediation
Individual boundary incidence	**B4 (B4_EXECUTION_TO_AUDIT)**	**247**	**42.3%**	**224**	**50.7%**	Hash-chained/Merkle provenance logs, execution receipts, and post-hoc anomaly reconstruction
Individual boundary incidence	**B1 (B1_INTENT_TO_DELEGATION)**	**172**	**29.5%**	**158**	**35.7%**	User intent compilation, consent-lattice refinement, and pre-ingestion prompt sanitization
Exact Boundary_Codes combination #1	**B2;B3** *(Delegation + Execution)*	**176**	**30.1%**	162	36.7%	Canonical tool-use pattern: metadata/context poisoning at B2 triggering unauthorized execution at B3, or paired B2+B3 gateways
Exact Boundary_Codes combination #2	**B2;B3;B4** *(Delegation + Exec + Audit)*	**90**	**15.4%**	89	20.1%	Runtime mediation proxies combining pre-execution policy checks (B2/B3) with tamper-evident logging (B4)
Exact Boundary_Codes combination #3	**B1;B2;B3;B4** *(Full 4-Boundary Stack)*	**80**	**13.7%**	78	17.6%	End-to-end governance architectures linking user intent commitments (B1) through execution (B3) to audit (B4)
Exact Boundary_Codes combination #4	**B1;B2;B3** *(Intent + Delegation + Exec)*	**66**	**11.3%**	64	14.5%	Task-scoped capability compilers (CapScope, ACE, ConLeash, A-VIP) and intent-to-action hijacking chains
Exact Boundary_Codes combination #5	**B3** *(Single Execution Gate)*	**55**	**9.4%**	50	11.3%	Isolated runtime enforcement hooks, enclave execution benchmarks, and deterministic tool-call filters
Exact Boundary_Codes combination #6	**B4** *(Single Audit / Provenance)*	**31**	**5.3%**	27	6.1%	Standalone witness/observability systems (AAE, PunkGo, Hearsay), watermarking, and forensic verifiers

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Table S7. Individual boundary incidence and exact boundary combinations. (continued)

Analysis dimension	Boundary code or exact combination	Claim count (n = 584)	Claim incidence (%)	Unique study count (n = 442)	Study incidence (%)	Dominant architectural role in the corpus
Exact Boundary_Codes combination #7	**B2** *(Single Delegation / Request)*	**25**	**4.3%**	21	4.8%	Pre-execution credential issuance, OAuth/DID discovery checks, and prompt-compression attacks (COMA)
Exact Boundary_Codes combination #8	**B3;B4** *(Execution + Audit)*	**20**	**3.4%**	20	4.5%	Verifiable execution traces (VET), kernel-level action commit logs, and TEE-attested execution receipts
Exact Boundary_Codes combination #9	**B2;B4** *(Delegation + Audit)*	**15**	**2.6%**	15	3.4%	Blockchain-anchored inter-agent communication (BlkMaC) and verifiable dataset/provenance credentials
Exact Boundary_Codes combination #10	**B1;B2** *(Intent + Delegation)*	**11**	**1.9%**	10	2.3%	User-consent negotiation interfaces and pre-request privacy redaction pipelines

Table S8. Lifecycle stage and non-exclusive boundary involvement.

Operational lifecycle stage (Lifecycle_Stage)	Total claims (n = 584)	Share (%)	Involves B1	Involves B2	Involves B3	Involves B4	Primary mechanisms and attack vectors
TOOL_INVOCATION	**361**	**61.8%**	135	330	345	142	MCP gateway mediation, capability tokens, indirect prompt injection (IPI), tool poisoning, and metadata manipulation
EXECUTION	**61**	**10.4%**	9	28	59	14	OS/kernel sandboxes (AgentBound, WASM, eBPF), TEE confidential execution (Opal, Omega), and evidence-before-effect sagas (Janus)
POST_EXECUTION_AUDIT	**45**	**7.7%**	1	8	12	44	Merkle/hash-chained logs (NovaFabric, PunkGo), TEE witnessing (ARXIV-S-0141), and harness-tampering audits (Hearsay)
INTER_AGENT_COMMS	**35**	**6.0%**	5	32	23	19	A2A authentication (BlockA2A, Lineage), cross-agent taint tracking, and inference-time steganographic covert channels (BAM)
AGENT_DISCOVERY	**26**	**4.5%**	7	26	11	10	MCP server registry drift (ARXIV-S-0274), AgentCard spoofing, preference manipulation (MPMA), and OAuth DCR abuse (ARXIV-S-0437)
NEGOTIATION_ORCHESTRATION	**19**	**3.3%**	5	17	18	10	Agentic commerce/payment mandates (AP2, A-VIP, A402), dynamic trust orchestration (TrustOrch), and BFT quorums (SQA)
PLANNING_REASONING	**14**	**2.4%**	3	14	9	2	Decoupled abstract planning (ACE, IPIGUARD, PlanTwin (G. Yu et al. 2026)) and lossy prompt-compression attacks (COMA)
MEMORY_PERSISTENCE	**14**	**2.4%**	2	11	10	6	Long-term memory poisoning (TMA-NM), cross-MCP memory exfiltration (MSA), and game-theoretic shared memory (EquiMem)
PROMPT_INGESTION	**9**	**1.5%**	5	6	4	0	Pre-inference PII redaction (AirGapAgent, LLM-Redactor) and GUI/accessibility-tree prompt injection

Table S9. Recorded formal-evidence types, implementation maturity, and within-scope support.

Formal evidence type (Formal_Evidence_Type)	Total claims (n = 584)	Share (%)	Tier 1 (n = 271)	Tier 2 (n = 313)	CONCEPTUAL / FORMAL_ONLY	PROOF_OF_CONCEPT / SIMULATOR	CONTROLLED_BENCHMARK	DEPLOYMENT_EVAL / PROD	SUPPORTED_WITHIN_SCOPE	PARTIALLY / NOT_SUPPORTED
INFORMAL_ARGUMENT	**201**	**34.4%**	78	123	14	121	54	12	149 (74.1%)	**52 (25.9%)**
NONE *(Empirical Only)*	**193**	**33.0%**	120	73	4	52	130	7	175 (90.7%)	**18 (9.3%)**
FORMAL_MODEL	**107**	**18.3%**	57	50	11	54	39	3	85 (79.4%)	**22 (20.6%)**
THEORETIC_PROOF	**54**	**9.2%**	9	45	3	30	18	3	48 (88.9%)	**6 (11.1%)**
TOOL_SUPPORTED_VERIFICATION	**15**	**2.6%**	2	13	1	12	2	0	11 (73.3%)	**4 (26.7%)**
SECURITY_REDUCTION	**11**	**1.9%**	4	7	1	7	2	1	11 (100.0%)	**0 (0.0%)**
LOGICAL_AXIOMS_WITH_PROOF	**3**	**0.5%**	1	2	0	2	1	0	3 (100.0%)	**0 (0.0%)**
Total across all claims	**584**	**100.0%**	**271**	**313**	**34 (5.8%)**	**278 (47.6%)**	**246 (42.1%)**	**26 (4.5%)**	**482 (82.5%)**	**102 (17.5%)**

Table S10. Non-exclusive limitation categories, evidence tiers, and support status.

Limitation code combination (Limitation_Codes_DG_AD_UG_NG)	All Claims (n = 584)	READY Claims (n = 530)	SUPPORTED_WITHIN_SCOPE (n = 482)	PARTIALLY / NOT_SUPPORTED (n = 102)	Primary methodological interpretation
AD *(Assumption-Dependent only)*	**207 (35.4%)**	**206 (38.9%)**	206	1	Guarantee holds within the author's stated model, but relies on material TCB assumptions (e.g., honest host/gateway, faithful policy compiler, uncompromised key custody)
AD;UG *(Assumption-Dependent + Unsupported Generalization)*	**134 (22.9%)**	**96 (18.1%)**	64	70	In addition to material TCB assumptions, prose claims generalize beyond small synthetic datasets, simulated primitives, or non-adaptive attacker evaluations
DG;AD *(Demonstrated Gap + Assumption-Dependent)*	**94 (16.1%)**	**94 (17.7%)**	90	4	Study demonstrates a concrete property failure or residual attack success rate (DG) alongside material TCB/deployment dependencies (AD)
DG;AD;UG *(All three limitation classes)*	**80 (13.7%)**	**67 (12.6%)**	56	24	Structural design/coverage gap (DG) + strong TCB dependency (AD) + overextended narrative claim (UG)
DG *(Demonstrated Gap only)*	**61 (10.4%)**	**61 (11.5%)**	61	0	Dedicated empirical attack or negative-result claim isolating a specific boundary failure (Stream A / secondary residual-failure claims)
DG;UG (5) and UG only (3)	**8 (1.4%)**	**6 (1.1%: 4 + 2)**	5	3	Empirical measurement or benchmark claim whose narrative generalization exceeds the tested sample denominator
Cumulative incidence (All 584 / READY 530)	**AD: 515 (88.2%) All / 463 (87.4%) READY**	**DG: 240 (41.1%) All / 226 (42.6%) READY**	**UG: 222 (38.0%) All / 169 (31.9%) READY**	**NG: 0 (0.0%) All & READY**	88.2% (515/584) All and 87.4% (463/530) READY claims depend on external TCB components; 95.1% (97/102) of PARTIALLY/NOT_SUPPORTED claims involve UG

Table S11. RQ3 metric units and source-audit/release dispositions.

Measurement unit (Unit)	Observation count (n = 1,111)	Share of RQ3 (%)	Initial SOURCE_ REPORTED	Initial SOURCE_ CONFLICT	Initial METHOD_ UNCLEAR	NOT_REPO RTED (NR omission)	Final Released (REPORTED + RESOLVED)	Held at EXTRACTED (FLAGGED / NR)
percent (%) *(ASR, blocking rate, utility, F1, overhead %)*	**374**	**33.7%**	337	9	28	0	**342 (91.4%)**	32 (8.6%)
ms *(milliseconds)* *(primitive, gate, and pipeline latency)*	**312**	**28.1%**	295	11	6	0	**275 (88.1%)**	37 (11.9%)
s *(seconds)* *(end-to-end task, LLM judge, FHE, consensus time)*	**142**	**12.8%**	138	3	1	0	**139 (97.9%)**	3 (2.1%)
tokens *(prompt/context overhead, token amplification)*	**125**	**11.3%**	122	1	2	0	**118 (94.4%)**	7 (5.6%)
fraction (0–1) *(precision, recall, DSR, FNR, leakage ratio)*	**96**	**8.6%**	95	0	1	0	**95 (99.0%)**	1 (1.0%)
NOT_REPORTED (NR) *(explicitly unmeasured runtime cost)*	**20**	**1.8%**	0	0	0	20	**0 (0.0%)**	20 (100.0%)
KB / bytes *(capsule, proof, public-key, signature size)*	**17**	**1.5%**	15	2	0	0	**17 (100.0%)**	0 (0.0%)
USD / gas *(API token cost, on-chain settlement cost)*	**12**	**1.1%**	11	1	0	0	**11 (91.7%)**	1 (8.3%)
ops/s / TPS / s/token *(throughput and generation rate)*	**13**	**1.2%**	11	1	1	0	**11 (84.6%)**	2 (15.4%)
Total RQ3 observations	**1,111**	**100.0%**	**1,024 (92.2%)**	**27 (2.4%)**	**40 (3.6%)**	**20 (1.8%)**	**1,008 (90.7%)**	**103 (9.3%)**

Table S12. Deduplicated eligible latency values by operational level and mechanism family.

Stratum (v3.5 Finalized Outcome-Independent & Sibling-Deduplicated; 0 HOLD)	All Dedup Obs. (n = 399) q25 / median / q75 of extracted values (ms)	READY Dedup Obs. (n = 362)	READY Dedup q25 / median / q75 of extracted values (ms) [IQR]	READY Canonical Reports (median of report medians)	Measurement interpretation & source-reported exemplars
L1: Primitive / Crypto Micro-Op	n = 90 0.22 / 1.80 / 13.41	**78**	0.26 / **1.76 ms** / 18.27 [IQR 18.01]	33 reports (**4.30 ms**)	Isolated signature, MAC, KEM, hash/Merkle commit, or certificate check (e.g., HBHC secp256k1 0.156 ms (Deochake 2026), PunkGo Merkle commit 1.26 ms (Zhang 2026), DCP HMAC-SHA256 on ESP32 15.6 ms (Yang 2026), and SecuAudit exact pairing ops 282–123,910 ms [REC-031..036])
L2: Per-Request Policy / Gate Check	n = 61 2.83 / 12.40 / 36.20	**51**	2.15 / **10.80 ms** / 32.55 [IQR 30.40]	32 reports (**11.15 ms**)	Synchronous per-tool-call reference monitor or gateway check (AgentSpec DSL 1.11–2.83 ms (H. Wang et al. 2026), Janus saga gate 5.39 ms (Arslan 2026), AEGIS-Router warm TEE 5.7 ms (Xie et al. 2026), ConLeash Datalog 8.2 ms (Y. Li, Y. Chen, et al. 2026), PayPal MCP auth cache hit 9.9 ms (H. Li et al. 2026), SAL mediation 12.4 ms)
L3: Session / Enclave / Consensus / Proof Setup	n = 42 10.04 / 262.88 / 8,375.00	**40**	12.42 / **262.88 ms** / 8,977.50 [IQR 8,965.08]	17 reports (**100.00 ms**)	Cold enclave RA-TLS boot (MemTrust 150–250 ms (X. Zhou et al. 2026)), smart-contract/BFT settlement (Poseidon HTLC 79.50–91.25 ms [REC-042/043], BlockA2A 64–135 ms (Zou et al. 2025)), Groth16 proof generation (351.9–1,970 ms (Cai et al. 2026; Llambí-Morillas and Fernández-Fernández 2026)), SQA quorum (4,120 ms (He and D. Yu 2026)), and CapSeal TFHE/LowMC (9,000–9,360,000 ms (Jin et al. 2026)); excludes 3 quarantined Niya mAI Table III conflict rows (p95 < median) and 1 duplicate (PILOT-MET-007)

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Table S12. Deduplicated eligible latency values by operational level and mechanism family. (continued)

Stratum (v3.5 Finalized Outcome-Independent & Sibling-Deduplicated; 0 HOLD)	All Dedup Obs. (n = 399) q25 / median / q75 of extracted values (ms)	READY Dedup Obs. (n = 362)	READY Dedup q25 / median / q75 of extracted values (ms) [IQR]	READY Canonical Reports (median of report medians)	Measurement interpretation & source-reported exemplars
L4: End-to-End Agent Workflow Runtime	n = 206 42.54 / 1,125.00 / 9,586.75	**193**	80.00 / **1,700.00 ms** / 10,900.00 [IQR 10,820.00]	96 reports (**432.65 ms**)	Complete single-turn or multi-turn agent workflow execution including LLM inference, network I/O, and tool-invocation pipelines (PolicyAgent 250–412 ms (Goel and Apuri 2026), ACE 11,730–20,950 ms (E. Li et al. 2025), AgenTEE 485,180 ms, CapChain live cell 1,182,000 ms [REC-019])
Family 1: Classical Crypto / Token Gate	n = 119 0.37 / 4.60 / 32.45	**110**	0.32 / **4.60 ms** / 43.52 [IQR 43.20]	55 reports (**7.30 ms**)	Classical capability tokens, digital signatures, MACs, and Merkle trees (5.4 μs broker token exchange (Dantuluri and Sundi 2026) to 15.6 ms ESP32 HMAC-SHA256 (Yang 2026))
Family 2: PQC / Quantum Primitives	n = 13 2.00 / 3.20 / 20.33	**3**	0.08 / **0.08 ms** / 10.21 [IQR 10.12]	2 reports (**10.21 ms**)	**Small-sample caveat (n = 3 READY across 2 reports):** QSC ML-KEM encap 0.082 ms, ML-DSA verify 0.080 ms (ARXIV-S-0140 (Bishwas et al. 2026)); MAGIQ sidecar handshake 20.33 ms (ARXIV-S-0057 (Avizheh et al. 2026)); 10 rows from CAESF (Palla 2026) and SLR-M-0244 (Sudarsan et al. 2026) are in FLAGGED P3
Family 3: Deterministic Policy / Kernel	n = 22 0.97 / 12.35 / 396.98	**18**	8.65 / **50.50 ms** / 838.00 [IQR 829.35]	14 reports (**300.95 ms**)	Spans fast deterministic policy hooks (Lean 4 checker 0.22 ms (Koomullil 2026), AgentSpec 1.11–2.83 ms (H. Wang et al. 2026), ConLeash 8.2 ms (Y. Li, Y. Chen, et al. 2026), LGA deterministic layers 18 ms (Ge 2026)) to full container/obligation sandboxes (250–4,960 ms)
Family 4: End-to-End Agent Workflows	n = 141 15.60 / 250.00 / 10,900.00	**130**	16.18 / **354.00 ms** / 13,450.00 [IQR 13,433.83]	62 reports (**172.50 ms**)	End-to-end tool and multi-agent workflows evaluated with non-runtime scan/attack confounders separated

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Table S12. Deduplicated eligible latency values by operational level and mechanism family. (continued)

Stratum (v3.5 Finalized Outcome-Independent & Sibling-Deduplicated; 0 HOLD)	All Dedup Obs. (n = 399) q25 / median / q75 of extracted values (ms)	READY Dedup Obs. (n = 362)	READY Dedup q25 / median / q75 of extracted values (ms) [IQR]	READY Canonical Reports (median of report medians)	Measurement interpretation & source-reported exemplars
Family 5: Blockchain / Consensus Ledger	n = 35 92.12 / 944.00 / 3,274.55	**32**	88.31 / **872.00 ms** / 4,152.90 [IQR 4,064.59]	11 reports (**600.00 ms**)	Smart-contract and BFT consensus settlement (BlockA2A 64–135 ms (Zou et al. 2025), Poseidon HTLC 79.50–91.25 ms [REC-042/043], S5-SHB 477.8 ms, SC-A2A on-chain grant anchor 8,350 ms)
Family 6: Hardware TEE / ORAM Enclave	n = 31 12.85 / 1,800.00 / 5,870.00	**31**	12.85 / **1,800.00 ms** / 5,870.00 [IQR 5,857.15]	6 reports (**297.94 ms**)	Warm TEE proxies operate in 5.7–39.8 ms (AEGIS-Router (Xie et al. 2026), Omega (Bodea et al. 2025)), whereas Opal TDX+ORAM (2,320 ms (Kaviani et al. 2026)), MemTrust (150–15,000 ms (X. Zhou et al. 2026)), and AgenTEE (485,180 ms) drive the observation median
Family 7: LLM-Judge / Quorum / FHE / ZKP	n = 38 98.69 / 1,110.40 / 8,992.50	**38**	98.69 / **1,110.40 ms** / 8,992.50 [IQR 8,893.81]	17 reports (**3,700.00 ms**)	Reflects the multi-second cost of synchronous LLM judging (LGA ~980 ms (Ge 2026), ConLeash extractor 4,690 ms (Y. Li, Y. Chen, et al. 2026)), multi-agent quorums (SQA 1,450–4,120 ms (He and D. Yu 2026)), Groth16 proving (351.9–1,970 ms (Cai et al. 2026; Llambí-Morillas and Fernández-Fernández 2026)), and FHE (9,000–9,360,000 ms (Jin et al. 2026)); includes GraphSAGE window detector (0.69 ms, REC-053) and excludes 3 quarantined NiyamAI Table III conflict rows

Table S13. READY latency sensitivity to restricting source statistics to mean/point values.

Level	All eligible READY records	Median across all recorded values (ms)	Mean/point records only	Median across mean/point values (ms)
L1	78	1.7645	75	1.82
L2	51	10.80	43	14.20
L3	40	262.88	39	173.86
L4	193	1,700.00	180	2,035.00

Table S14. PQC migration stages at the claim and dependency-record levels.

Post-quantum migration stage (PQC_Migration_Stage)	Tier 1 claims (n = 271)	Tier 2 claims (n = 313)	Total claims (n = 584)	Claim share (%)	Linked RQ4 dependency records (n = 283)	Operational readiness definition and corpus status
NOT_DISCUSSED	257	151	**408**	**69.9%**	153	Recorded absence-of-discussion label; analyzed jointly with Stage 0 because batch coding differed
STAGE_0_PRE_QUANTUM_ONLY	2	150	**152**	**26.0%**	107	Recorded classical-only label; batch-dependent use, analyzed jointly with NOT_DISCUSSED (the main article)
STAGE_1_SUGGESTED_MIGRATION	7	4	**11**	**1.9%**	12	Acknowledges quantum vulnerability in prose and suggests ML-DSA, ML-KEM, SPHINCS+, or ZK-STARKs as future work
STAGE_2_ANALYTICAL_PROPOSAL	2	0	**2**	**0.3%**	3	Proposes a high-level post-quantum or hybrid architecture (ContextGuard-PQC, TAEA) without code or valid measurements
STAGE_3_SIMULATION_OR_EMULATION	1	3	**4**	**0.7%**	2	Simulates PQC or quantum-resilient behavior in software (e.g., AITH simulating ML-DSA-87 via HMAC-SHA256)
STAGE_4_PRIMITIVE_IMPLEMENTED	2	0	**2**	**0.3%**	4	Integrates standard PQC primitives (liboqs Kyber/Dilithium) in a standalone or simulated enterprise testbed
STAGE_5_AGENT_WORKLOAD_BENCHMARK	0	5	**5**	**0.9%**	2	Benchmarks real NIST PQC primitives (ML-KEM-768, ML-DSA-65, XMSS) inside an active multi-agent/MCP workload (MAGIQ, QSC)
STAGE_6_OPERATIONAL_DEPLOYMENT	0	0	**0**	**0.0%**	0	Production deployment of post-quantum cryptography in a live agentic system (**0 studies in the corpus**)
Total across corpus	**271**	**313**	**584**	**100.0%**	**283**	**95.9% (560/584) of claims are NOT_DISCUSSED or STAGE_0; only 24 claims (4.1%) reach STAGE_1–STAGE_5**

Table S15. Cryptographic dependency classes, exposure provenance, and linked populations.

Finalized v3.5 construction category & exposure certainty tier	All RQ4: Rec (Clm / Rep)	READY RQ4: Rec (Clm / Rep)	Record Share of Total (283) / Active (266)	Primary security functions across B1–B4	Construction-level interpretation and migration constraints
Tier A-1 (Cat 3): Explicit Classical Elliptic-Curve Signatures / Key Agreement (Ed25519, ECDSA, X25519)	**42** (40 / 40)	**41** (39 / 39)	14.8% / 15.8%	Agent manifests (.agent), AP2 payment mandates (B1/B2), execution receipts (B3/B4), and blockchain accounts (DEP-074 AgentFacts)	Direct exposure when the named classical construction supplies the relevant function. X25519 is key agreement, not a signature: evaluate ML-KEM for establishment and ML-DSA/SLH-DSA for signatures (NIST 2024a,b,c). A family label does not specify all parameters.
Tier A-2 (Cat 7): Pairing & Discrete-Log Zero-Knowledge Proofs (Groth16, PLONK-KZG, Halo 2KZG, Bulletproofs)	**22** (20 / 19)	**19** (17 / 16)	7.8% / 8.3%	Private policy compliance (CVA (Llambí-Morillas and Fernández-Fernández 2026)), verifiable proxy re-encryption (VF-RABPRE (Cai et al. 2026)), and confidential audit bridges (Sentinel DEP-113)	The recorded DLog/pairing assumption is quantum-exposed. Hash-based or lattice-based alternatives require separate soundness, privacy, parameter and implementation analysis; they are not interchangeable drop-in replacements.
Tier A-3 (Cat 8): Classical RSA & Bilinear-Pairing / Threshold Schemes (RSA-2048, BLS12-381, IBE, BBS+)	**18** (18 / 18)	**17** (17 / 17)	6.4% / 6.8%	Multi-agent quorum signatures (SQA (He and D. Yu 2026), BlockA2A (Zou et al. 2025)), identity-based encryption, and encrypted equality (BT-PKEET)	Function-specific migration is required: RSA encryption/signatures, pairing-based IBE, and aggregate signatures serve different purposes. Session MACs do not preserve public verifiability or threshold-signature semantics.
Tier B (Cat 2): Conditional Identity, Delegation & Transport Protocol Wrappers (OAuth 2.0/2.1, JWT/JWS, DID/VC, mTLS/SPIFFE)	**33** (33 / 33)	**25** (25 / 25)	11.7% / 12.4%	Principal-to-agent delegation tokens (B1/B2), MCP gateway admission (B3), and A2A identity (DEP-010, DEP-031)	Conditional on the actual signing/ key-establishment suite and certificate chain. Protocol names alone do not establish a classical default. FIPS 203 specifies ML-KEM, not a complete hybrid TLS profile.
Tier C (Cat 5): Conditional Hardware Enclaves & Silicon Attestation Roots (SGX/ TDX, SEV-SNP, Nitro, CCA, TPM 2.0)	**18** (18 / 17)	**15** (15 / 14)	6.4% / 6.8%	Confidential router/memory execution (B3) and attested audit witnesses (B4: Opal (Kaviani et al. 2026), AEGIS-Router (Xie et al. 2026), MemTrust (X. Zhou et al. 2026))	Conditional on the provisioned attestation and endorsement chain. Binding a PQ public key into attestation data does not upgrade a classical attestation signature or prove the vendor root quantum-resistant.

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Table S15. Cryptographic dependency classes, exposure provenance, and linked populations. (continued)

Finalized v3.5 construction category & exposure certainty tier	All RQ4: Rec (Clm / Rep)	READY RQ4: Rec (Clm / Rep)	Record Share of Total (283) / Active (266)	Primary security functions across B1–B4	Construction-level interpretation and migration constraints
Grover-1 (Cat 1): Cryptographic Hashes, Merkle Trees & Hash-STARKs (SHA-256, BLAKE3, RFC 6962, zk-STARK)	**36** (36 / 36)	**32** (32 / 32)	12.7% / 13.5%	Tamper-evident audit logs (B4: PunkGo (Zhang 2026), Semantic Gateway DEP-089), manifest digests (B2), and hash-based proofs (DEP-037, DEP-075)	Retain the source construction and examine the property needed: preimage, collision, commitment or proof soundness. Output length alone does not establish a common post-quantum strength for all these constructions.
Grover-2 (Cat 4): Symmetric MACs & Capability Tokens (HMAC-SHA256, Macaroons, PRF/HKDF)	**24** (24 / 24)	**22** (22 / 22)	8.5% / 9.0%	Fast per-hop capability attenuation and source-channel binding (B2/B3: DCP DEP-120 (Yang 2026), TMA-NM (Louck 2026))	Key-search, tag length, query bounds and key distribution matter. Symmetric authentication may support scoped tickets, but shared-key verifiers can also create tags; public verifiability is a separate requirement.
Grover/Lattice-3 (Cat 6): Symmetric Encryption, ORAM, MPC & Lattice FHE (AES-256-GCM, TFHE, CKKS, Shamir)	**13** (11 / 11)	**9** (7 / 7)	4.6% / 4.9%	Long-term memory confidentiality (Opal (Kaviani et al. 2026)), operator-blind secret derivation (CapSeal (Jin et al. 2026)), and privacy-preserving inference (DEP-021, DEP-026, DEP-114)	This historical group combines different assumptions. ORAM and MPC are not primitive-level quantum certifications; examine their encryption, channels, parameters and adversary models separately.
PQC (Cat 9): Evaluated or Proposed Post-Quantum & Quantum Constructions (ML-KEM, ML-DSA, XMSS, QRNG, QKD)	**16** (11 / 11)	**8** (6 / 6)	5.7% / 6.0%	Post-quantum agent governance (MAGIQ (Avizheh et al. 2026)), sidecar PQC-TLS 1.3 (QSC (Bishwas et al. 2026)), and hybrid PKI proposals (CAESF (Palla 2026), ContextGuard (Gentyala et al. 2026))	This category includes proposals and quantum technologies as well as implemented PQC. QRNG or simulated QKD does not establish post-quantum authentication of an agent workflow; consult the migration stage and assessed scope.
Unspecified (Cat 10a-i): Unspecified Algorithm / Generic Signed, ZKP or HE Scheme (Parameters_As_Reported = NR)	**44** (44 / 44)	**36** (36 / 36)	15.5% / 16.5%	Generic signed governance/ audit records, permissioned chaincode, generic ZKP/HE claims (40 initial + 4 finalized FINAL_QUALIFY_SUITE_NR: DEP-035, 048, 051, 095)	**Unspecified (Suite = NR):** Primary study invokes a cryptographic mechanism without reporting concrete curve/suite parameters in the PDF; requires explicit algorithm pinning

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Table S15. Cryptographic dependency classes, exposure provenance, and linked populations. (continued)

Finalized v3.5 construction category & exposure certainty tier	All RQ4: Rec (Clm / Rep)	READY RQ4: Rec (Clm / Rep)	Record Share of Total (283) / Active (266)	Primary security functions across B1–B4	Construction-level interpretation and migration constraints
Subtotal: Active Adjudicated Cryptographic & Hardware-Attestation Dependencies (Cats 1–9 + 10a-i)	**266** (231 / 228)	**224** (197 / 195)	94.0% / 100.0%	**222 identified construction-class records (189 clm / 186 rep; 188 READY across 162 clm / 160 rep) + 44 Suite = NR records (44 clm / 44 rep; 36 READY across 36 clm / 36 rep)**	**133 Shor-exposed records (82 Tier A [76 clm / 75 rep; 77 READY across 72 clm / 71 rep] + 33 Tier B + 18 Tier C) = 50.0% of 266 active / 59.9% of 222 concrete**
Excluded Non-Crypto (Cat 10b + 10a-ii): Non-Crypto Baselines (2), Excluded Own Non-Crypto (8), Rejected Labels (7)	**17** (17 / 17)	**17** (17 / 17)	6.0% / – *(Excluded)*	2 explicit baselines (DEP-0005/0006), 8 FINAL_EXCLUDE_OWN_CRYPTO (DEP-090, 092, 097, 116, 118, 119, 121, 122), and 7 FINAL_REJECT_CLASS (DEP-091, 100, 102, 104, 105, 108, 123)	**Not Applicable (EXCLUDED_*):** Non-cryptographic guardrails, scanners, or caches (1 Tier 1 + 16 Tier 2 reports) excluded from the 266 active cryptographic denominator
Total Finalized RQ4 Inventory (0 HOLD)	**283** (248 / 245)	**241** (214 / 212)	**100.0% (266 Active + 17 Excluded)**	**All B1–B4 security functions across 248 unique claims (214 READY + 34 FLAGGED) and 245 unique reports (212 with READY + 34 with FLAGGED, 1 overlap)**	**133 Shor-exposed (47.0% of 283 / 50.0% of 266 active) + 73 Grover/Lattice (67 clm / 67 rep) + 16 PQC (11 clm / 11 rep) + 44 Suite=NR + 17 Excluded Non-Crypto**

Table S16. Descriptive sensitivity across the peer-reviewed and preprint evidence tiers.

Synthesis indicator (RQ1–RQ4)	Tier 1 Only (n = 212 Peer-Reviewed; 271 Claims)	Tier 2 Only (n = 230 arXiv Preprints; 313 Claims)	Combined Full Corpus (n = 442 Studies; 584 Claims)	Difference (Tier 1 minus Tier 2; percentage points for rates)	Sensitivity conclusion & structural interpretation
Mean claims per study	1.28 (271/212)	1.36 (313/230)	1.32 (584/442)	-0.08	**Descriptive similarity:** Comparable claim granularity across both tiers
RQ1: MULTI_BOUNDARY claim share	**79.7%** (216/271)	**81.8%** (256/313)	**80.8%** (472/584)	**-2.1 pp**	**Descriptive similarity:** Multi-boundary involvement (~80%–82%) holds across both peer-reviewed venues and preprints
RQ1: B2 (Delegation → Request) incidence	**81.2%** (220/271)	**80.5%** (252/313)	**80.8%** (472/584)	**+0.7 pp**	**Descriptive similarity:** B2 incidence is high in each tier; universality is not established
RQ1: B3 (Request → Execution) incidence	**85.2%** (231/271)	**83.1%** (260/313)	**84.1%** (491/584)	**+2.1 pp**	**Descriptive similarity:** B3 execution gating is the primary focal point in both tiers
RQ1: B1 (Intent → Delegation) incidence	15.1% (41/271)	41.9% (131/313)	29.5% (172/584)	-26.8 pp	**Complementarity:** 2026 Tier 2 preprints more frequently model explicit B1 user-intent/consent lattices (ConLeash, A-VIP)
RQ1: B4 (Execution → Audit) incidence	32.8% (89/271)	50.5% (158/313)	42.3% (247/584)	-17.7 pp	**Complementarity:** Recent Tier 2 studies expand post-execution provenance and witness auditing (NovaFabric, Hearsay)
RQ2: SUPPORTED_WITHIN_SCOPE rate	**81.9%** (222/271)	**83.1%** (260/313)	**82.5%** (482/584)	**-1.2 pp**	**Descriptive similarity:** The observed support rates are similar; peer review effects are not identified (18.1% Tier 1 vs. 16.9% Tier 2 PARTIALLY/NOT_SUPPORTED)
RQ2: Final Claim_QA = READY release rate (Paket B)	**89.7%** (243/271)	**91.7%** (287/313)	**90.8%** (530/584)	**-2.0 pp**	**Descriptive similarity:** High post-adjudication release rate across both tiers (28 Tier 1 and 26 Tier 2 claims retained at FLAGGED)
RQ2: Focused formal-evidence category share	2.6% (7/271)	7.0% (22/313)	5.0% (29/584)	-4.4 pp	**Focused category share below 8% in each tier:** Tier 2 captures recent Lean 4, TLA+, Tamarin, ProVerif, and Alloy artifacts

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Table S16. Descriptive sensitivity across the peer-reviewed and preprint evidence tiers. (continued)

Synthesis indicator (RQ1–RQ4)	Tier 1 Only (n = 212 Peer-Reviewed; 271 Claims)	Tier 2 Only (n = 230 arXiv Preprints; 313 Claims)	Combined Full Corpus (n = 442 Studies; 584 Claims)	Difference (Tier 1 minus Tier 2; percentage points for rates)	Sensitivity conclusion & structural interpretation
RQ2: Assumption-dependent (AD) incidence	77.1% (209/271)	97.8% (306/313)	88.2% (515/584)	-20.7 pp	**Directionally consistent (>77% in both tiers):** TCB assumptions bound the vast majority of claims in both tiers
RQ3: Benchmark / PoC / Deployment maturity share	**77.1%** (209/271)	**91.1%** (285/313)	**84.6%** (494/584)	-14.0 pp	**High empirical grounding in both tiers** (reflecting the mandatory Tier 2 preprint rigor gate in the main article)
RQ4: Reports with RQ4 inventory / active crypto deps	**55.2% (117/212 inv.) / 54.7% (116/212 active)**	**55.7% (128/230 inv.) / 48.7% (112/230 active)**	**55.4% (245/442 inv.) / 51.6% (228/442 active)**	**-0.5 pp (inv.) / +6.0 pp (active)**	**Consistent:** 245/442 reports (248 claims, 283 records) have RQ4 entries; after excluding 17 non-crypto rows (1 Tier 1 + 16 Tier 2), 228/442 (51.6%; 231 claims, 266 records) have active crypto dependencies
RQ4: Pre-quantum or unaddressed PQC (NOT_DISCUSSED + STAGE_0)	**95.6%** (259/271)	**96.2%** (301/313)	**95.9%** (560/584)	**-0.6 pp**	**Descriptive similarity:** The harmonized no-demonstrated-migration share is similar in Tier 1 (4.4% Stage 1–4) and Tier 2 (3.8% Stage 1–5)

Table S17. Claim-role contrasts, with exact denominators for all and READY populations.

Claim role / QA view	Claims	Reports	Supported within scope	DG	UG	Focused formal types	Including THEORETIC_PROOF
Mechanism/framework/prototype, All	346	328	256 (74.0%)	81 (23.4%)	141 (40.8%)	26 (7.5%)	73 (21.1%)
Mechanism/framework/prototype, READY	298	285	256 (85.9%)	69 (23.2%)	94 (31.5%)	25 (8.4%)	71 (23.8%)
Attack/benchmark/attack-surface, All	238	218	226 (95.0%)	159 (66.8%)	81 (34.0%)	3 (1.3%)	10 (4.2%)
Attack/benchmark/attack-surface, READY	232	214	225 (97.0%)	157 (67.7%)	75 (32.3%)	2 (0.9%)	9 (3.9%)

Table S18. Evidence-tier contrasts within mechanism, framework, and prototype claims.

Tier	Claims	Within-scope support	DG	UG	Focused formal types
Tier 1	183	137 (74.9%)	8 (4.4%)	51 (27.9%)	7 (3.8%)
Tier 2	163	119 (73.0%)	73 (44.8%)	90 (55.2%)	19 (11.7%)

Table S19. Cross-boundary evidence, warranted inferences, and remaining obligations.

Boundary obligation	Evidence in the corpus	Defensible synthesis	What remains unestablished
B1: represented task to delegated authority	A-VIP distinguishes signed transaction fields from selection semantics; ConLeash depends on extracted predicates (Y. Li, Y. Chen, et al. 2026; Louck et al. 2026)	Approval should bind an explicit representation, with ambiguity and escalation handled outside untrusted content	Semantic fidelity of unrestricted natural language; reliable approval under deception or fatigue
B2: authority to a current request	Scope, origin, quota and expiry mechanisms in CapScope, TMA-NM and SAGA (Bouras et al. 2026; Louck 2026; Syros et al. 2025)	Request evaluation must bind the relevant principal, object, operation, scope and policy state	Benign behavior within valid permissions; complete revocation during partitions; universal injection resistance
B3: checked request to observed effect	PCC-LLM exposes omitted environmental predicates; AEGIS-Router distinguishes model and implementation configurations (Koomullil 2026; Xie et al. 2026)	The checked representation must correspond to the executed arguments, code and relevant mutable state	Complete mediation across arbitrary tools, irreversible remote actions and unmodeled state
B4: effect to verifiable evidence	Hearsay distinguishes valid logs from complete capture; witnessing appraisals retain recorder and implementation assumptions (Dai et al. 2026; Rowstron 2026)	Integrity, authenticity, ordering and capture coverage need separate claims and tests	Truth of self-reported events, completeness outside the observation surface, and guaranteed availability

Table S20. Candidate design requirements, falsification tests, and residual trust.

Requirement	Falsification test	Residual assumption / operational trade-off
I. Bind approval to an explicit task/policy representation (B1)	Change merchant, resource, purpose or high-consequence parameters between approval and invocation; test ambiguous requests and misleading approval displays	A trusted presentation and approving principal are required. Deterministic checking cannot recover unstated intent. Human approval adds delay and can fail.

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Table S20. Candidate design requirements, falsification tests, and residual trust. (continued)

Requirement	Falsification test	Residual assumption / operational trade-off
II. Bind delegated authority to each request (B2)	Test attenuation, recipient binding, replay, expired/revoked grants, policy-version changes and cross-tenant handle substitution	Trusted issuers, key custody and synchronized state remain relevant. Blind handles reduce secret exposure but do not prevent authorized misuse.
III. Bind authorization to execution (B3)	Mutate arguments or resource state after checking; exercise alternate APIs, retries, crashes and partial remote effects	Complete adapter coverage and race handling are required. A write-ahead record does not make an external operation atomic or guarantee rollback.
IV. Preserve provenance through feedback and memory (B2/B3)	Inject instructions through return values, exceptions, cached state and changed tool manifests; test benign-data corruption by filters	Typed envelopes and hashes identify structure/version, not semantic harmlessness. Sanitization can lose useful information and requires evaluation.
V. Make capture coverage and failure visible (B4)	Omit, reorder, duplicate or truncate records; fork histories; disconnect the witness; crash between record commit and tool effect	The witness has its own TCB. Record uncertain completion explicitly; checkpoints help expose forks only under the assumed verifier/communication model.

Table S21. Function-specific migration decisions and required supporting evidence.

Security function	Migration decision / candidate	Required evidence and unresolved constraint
Confidential channels and stored secrets	Identify classical key establishment and evaluate authenticated PQ/hybrid establishment with suitable symmetric encryption	Separate secrecy lifetime from session duration; measure handshake, resumption, negotiation failures and deployment compatibility
Mandates, delegation and cross-domain receipts	Evaluate PQ signatures while preserving issuer, audience, scope, freshness and policy-version binding	Establish whether independent verification is required. A shared-key MAC changes the verifier's ability to forge and does not replace a public signature without a changed trust model
Logs and long-lived evidence	Analyze capture, checkpoint authenticity, hash/proof assumptions and evidence renewal together	Define the property and security strength sought. SLH-DSA is stateless; XMSS/LMS require state management and prevention of one-time-key reuse (Cooper et al. 2020; NIST 2024c)
Zero-knowledge compliance and aggregation	Identify the exact exposed commitment/ proof assumptions before assessing alternatives	Re-evaluate privacy, soundness, setup, proof size and verification semantics. FIPS 203–205 do not standardize a replacement for every proof or aggregate-signature system
Hardware-backed attestation	Inventory the endorsement/attestation chain and bind session keys to the intended attested identity and measurements	A PQ key bound by a classical quote still depends on classical authentication of that binding. Vendor-specific trust-chain migration and rollback/ revocation policy remain separate obligations

Table S22. Research priorities and the minimum evidence needed to evaluate progress.

Priority	Minimum evidence	What it would resolve
Claim and assumption specification	Protected object, attacker capabilities, trusted components, enforcement location, scope and explicit non-goals	Whether an attack and a defense evaluate the same property
Model-to-code correspondence	Exact theorem/property, checker and version, assumptions/axioms, model bounds, runnable artifact, and links to implementation paths	Whether a tool name denotes an asserted result, bounded check, or independently replayable proof
Complete-path security and utility	Matched tasks/backends; adaptive and cross-channel attacks; confusion counts with positive and negative cases; failed runs and benign completion	Whether protection transfers beyond the evaluated examples and at what utility cost
Performance estimand and uncertainty	Timer start/stop, baseline, setup frequency, configuration, sample count, source statistic, uncertainty and raw repetitions where possible	Whether a timing is total runtime, incremental overhead, a modeled quantity or a component benchmark
Capture and recovery	Explicit event coverage, effect acknowledgments, bypass/fork tests, crash/replay/revocation experiments	Whether valid audit evidence corresponds to the relevant execution history
Migration fidelity	Exact primitive/library/parameter set, real versus simulated components, security function, artifact lifetime and complete serialized sizes	Whether a PQC label denotes implemented migration and which residual classical dependencies remain

3 Detailed RQ visualizations

The figures retain their source-specific populations and interpretation limits. Colors are accompanied by counts, labels or tabular counterparts. Their numeric inputs reproduce the original 55-row summary; only embedded legacy figure numbers were removed for the JAIR layout.



Fig. S1. Boundary involvement across claim/report populations and claim roles; exact combinations and lifecycle mapping. Involvement does not by itself establish protection or attack propagation.

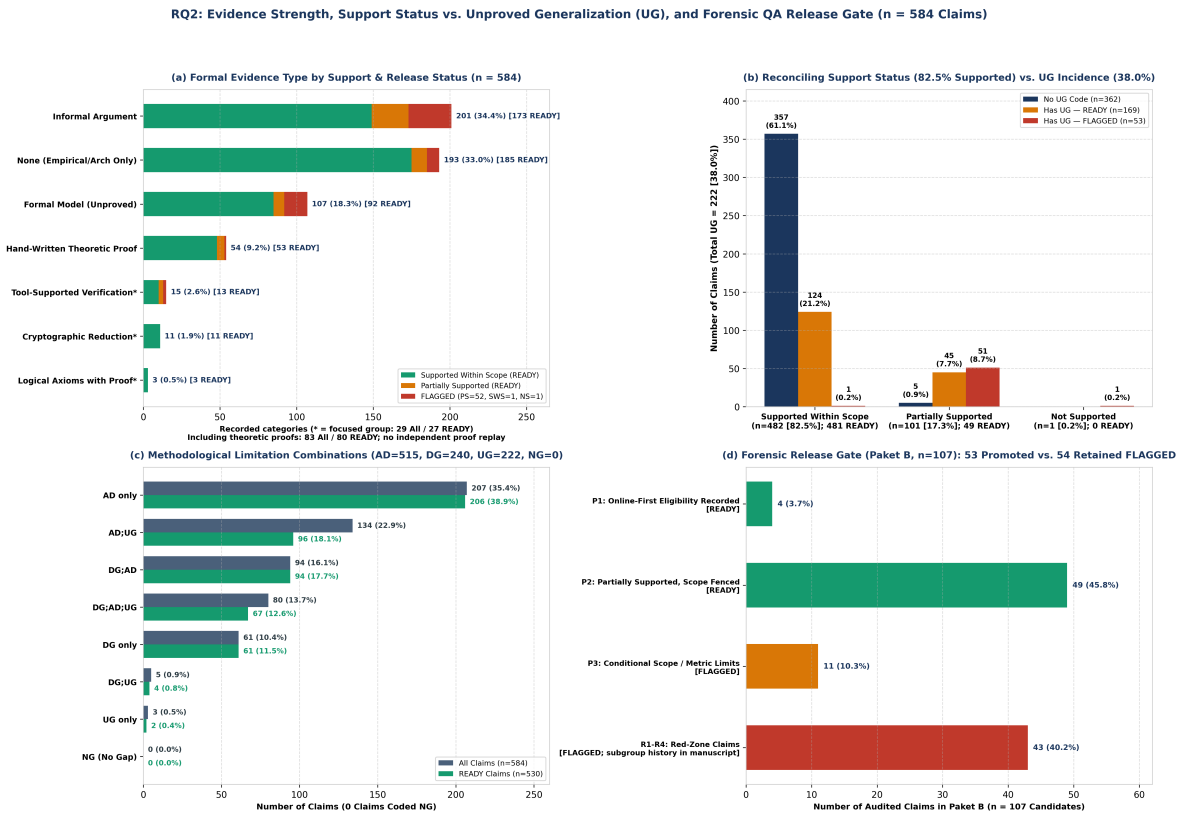


Fig. S2. Recorded formal-evidence categories, support, limitations, and QA candidate groups. The starred grouping contains 29 claims; including theoretical proofs yields 83. Category membership is not independent proof verification. Panel d groups red-zone R1–R4 together because their separate history is not encoded in the distributed disposition field.

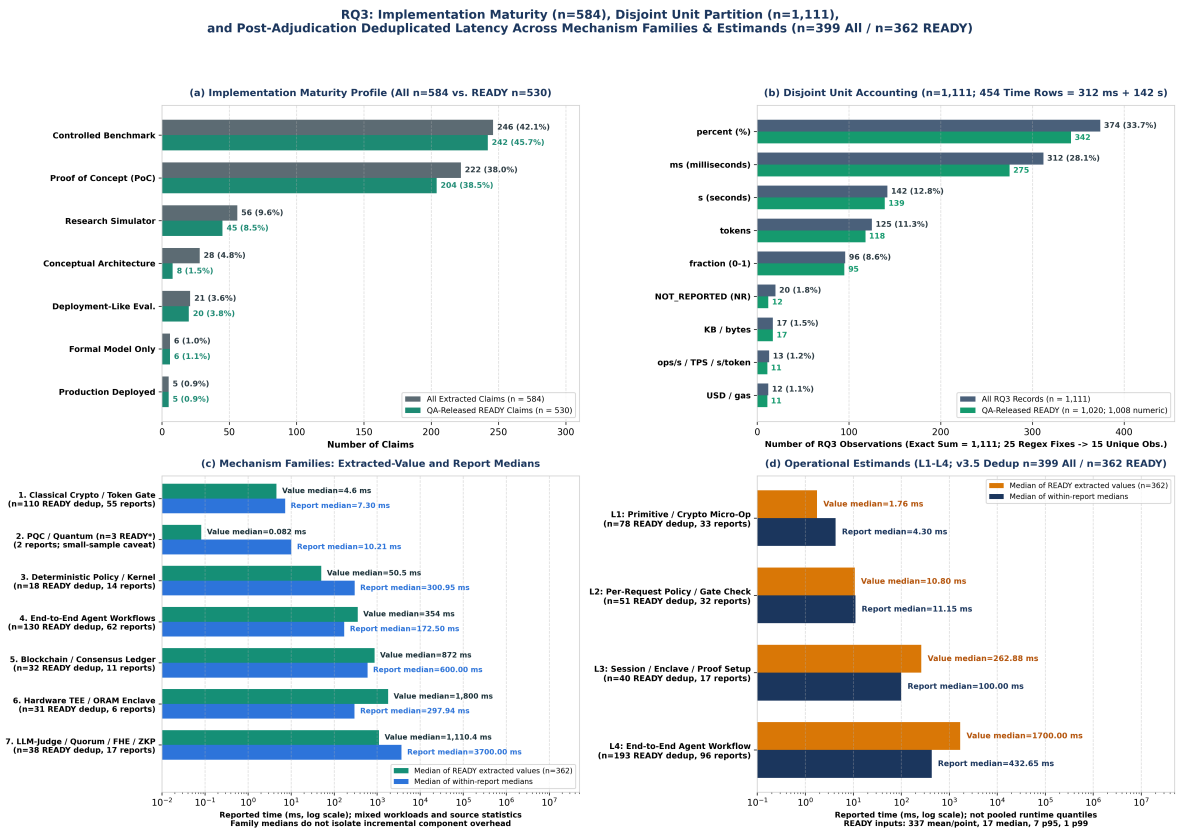


Fig. S3. Maturity, disjoint metric units, and final eligible latency strata. Panels c–d summarize extracted values and medians of within-report medians. Source statistics are heterogeneous: 337 mean/point values, 17 medians, seven p95 and one p99 among 362 READY records. These are not pooled runtime quantiles or incremental overhead estimates.

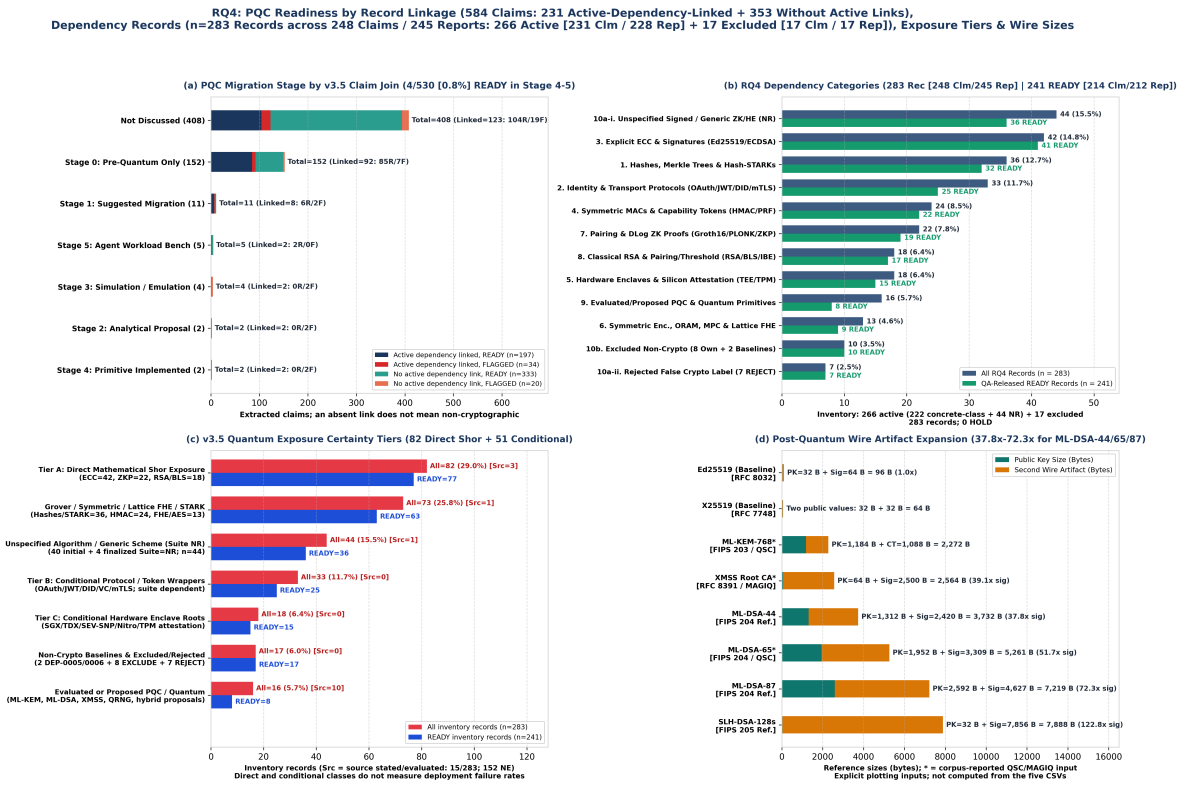


Fig. S4. PQC migration evidence, active dependency classes and exposure provenance, with explicitly labeled reference sizes for selected signature constructions. Dependency records, claims and reports are distinct denominators; reference-size inputs in panel d are not pooled measurements from the corpus.

A Search provenance and full expressions

The expressions below preserve the reported search history and its qualifications. ACM interface settings were not independently recovered. Search-log hashes identify the recorded artifacts; they do not establish their contents independently.

Table S23. Search and export log, including reported artifact identifiers and checksums.

Information source / platform	Executed date (UTC)	Query / export identifier	Raw records retrieved	Net unique records (post-filter / dedup)	Immutable artifact filename & SHA-256 checksum
Scopus (Advanced Search, `TITLE-ABS-KEY, AND NOT DOCTYPE(cr OR bk)`)	2026-10-15	S1 (Crypto, TEE & PQC)	137	136	`export_S1.csv` (28ea4a786109520258629150 9d21c8d171c5f22a16be ae5e14788d0733140839)
Scopus (Advanced Search, `TITLE-ABS-KEY, AND NOT DOCTYPE(cr OR bk)`)	2026-10-15	S2 (Delegation, Identity & Auth)	70	70 (+46 inc.)	`export_S2.csv` (fa59 70f5bce19c673be67079 95f342aaa8dcb6aeaea6 8292d5dc7d98f607079f)
Scopus (Advanced Search, `TITLE-ABS-KEY, AND NOT DOCTYPE(cr OR bk)`)	2026-10-15	S3 (Boundary Attacks & MCP)	103	103 (+75 inc.)	`export_S3.csv` (782b 3ba62bc60b281fd0e5b7 1da7a2ee299cc23a91fc 76e8728c342766c6c476)
Scopus Consolidated	2026-10-15	S1–S3 Union (SCOPUS-S-0001..0257)	310	**257**	`Scopus_S1_S2_S3_COMBINED_DEDUP.csv` (d6cf 515d9b82035277e95a6f ac840f782b2dd3822549 c17bd8d04ed4498d5d50)
IEEE Xplore (Command Search, `All Metadata`, no UI facet filters)	2026-10-15	I1a (Crypto, Attestation & TEE)	78	78	`export_I1a.csv` (e77d 603d467262dfd9283b14 596acb758bc475bdf30b f9e87bba43f510424d7c)
IEEE Xplore (Command Search, `All Metadata`, no UI facet filters)	2026-10-15	I1b (FHE, MPC & PQC)	43	43 (+24 inc.)	`export_I1b.csv` (0ef8 0198e4d63fe1d39fc2be 3b51e8834965dd6c5935 062b34f44639e33b9d3b)
IEEE Xplore (Command Search, `All Metadata`, no UI facet filters)	2026-10-15	I2 (Delegation, Identity & Auth)	55	55 (+39 inc.)	`export_I2.csv` (bbf8 8741539cfd1e24d823e2 dd1311d5cad6eabb1b80 c21b8699df839c6198b6)

Continued on next page

Table S23. Search and export log, including reported artifact identifiers and checksums. (continued)

Information source / platform	Executed date (UTC)	Query / export identifier	Raw records retrieved	Net unique records (post-filter / dedup)	Immutable artifact filename & SHA-256 checksum
IEEE Xplore (Command Search, `All Metadata`, no UI facet filters)	2026-10-15	I3a (Agent Boundary Attacks)	22	22 (+12 inc.)	`export_I3a.csv` (7753 1d54c1266b0864b9ed0b 587524844439647dce7d 9f4f20ac7473f644a6d2)
IEEE Xplore (Command Search, `All Metadata`, no UI facet filters)	2026-10-15	I3b (MCP Security & Runtime Policy)	58	58 (+33 inc.)	`export_I3b.csv` (2ff6 8b3bb96f652c46f92834 3e022ede138f1e650249 f84c58b5aeba2c006dc0)
IEEE Xplore Consolidated	2026-10-15	I1a–I3b Union (IEEE-S-0001..0186)	256	**186** (101 Scopus overlap; **+85 new**)	`IEEE_I1a_I1b_I2_I3a_I3b_COMBINED_DEDUP.csv` (7dc4a0160ab4ebc9 f7d31946c55533c3fde9 3076cd5d71ee80b70b1b 6749010a)
Scopus + IEEE Master	2026-10-15	SLR-M-0001..0342	566	**342**	`Master_Scopus_IEEE_COMBINED_DEDUP.csv` (8d21acd7bafd805be172 63bdccf5b62146607812 6fa5a1b40ea050be44b9 a684)
ACM Digital Library (Advanced Search BibTeX Exports)	2026-10-15	ACM-S1 (`acm.bib`: 771), ACM-S2 (`acm(1).bib`: 1,000), ACM-S3 (`acm(2).bib`: 381)	2,152	**113** TAK-verified (23 Scopus/IEEE overlap; **+90 new**)	`acm.bib` (26d5f7d8...), `acm(1).bib` (bafc9f84...), `acm(2).bib` (ab1ed454...) → `ACM_S1_S2_S3_FILTERED_DEDUP.csv` (1190a1ebca7311607385 c94d0c0072237cb8873c ab51bfe3b58fff9ff093 6eda)

Continued on next page

Table S23. Search and export log, including reported artifact identifiers and checksums. (continued)

Information source / platform	Executed date (UTC)	Query / export identifier	Raw records retrieved	Net unique records (post-filter / dedup)	Immutable artifact filename & SHA-256 checksum
arXiv Official Atom API (export.arxiv.org/api/query, cat:cs.CR/AI/MA/SE/DC)	2026-10-15T15:15:09Z	ARXIV_S1a (3,814), ARXIV_S1b (76), ARXIV_S2 (145), ARXIV_S3 (107)	4,142	**529** passing gate (35 superseded by Tier 1; **+494 new**)	export_ARXIV_ALL_RAW.csv (e58f5de0eb711011199e7b38b8200bc27e251e5f304ba1199fba9332fab247cf) → ARXIV_S1_S2_S3_FILTERED_DEDUP.csv (8ba9feeb852347782e1ed721bc12abe52ff0706e273fbdf1d22d9b5588988f7d)
Unified Master Corpus (Tier 1 + Tier 2)	2026-10-15T15:15:09Z	Consolidated Pre-Screening Corpus	**6,860**	**926** (432 Tier 1 + 494 Tier 2)	Master_SLR_FULL_CORPUS_DEDUP.csv (fe9786f7fc40ce3f2e9ab54003f1d7f0e3ea86a315521ca3b9d123b70d18464a)
Backward / Forward Citation Chasing	Not executed	10-seed check only (no snowballing)	0	0	Omitted prior to 15 Oct 2026 extraction lock; reported as a search limitation in Section~/ref{sec:4.5}

Preserved from manuscript reporting release 3.6.3. Search history and provenance qualifications remain applicable.

A.1 Executed source-specific expressions

To document the executed searches under PRISMA-S (Rethlefsen et al. 2021), the supplied verbatim Scopus/IEEE search expressions and the available ACM/arXiv query summaries for **15 October 2026** are recorded below:

A.1.1 Scopus Advanced Search (S1–S3).

- **S1 (Cryptographic, Hardware-Root & Post-Quantum Assurance — 137 raw rows; 136 unique):**
  TITLE-ABS-KEY(
  ("LLM agent*" OR "agentic AI" OR "agentic LLM*" OR "agentic workflow*" OR
  "large language model agent*" OR "LLM-based agent*" OR "LLM-powered agent*" OR
  "LLM-driven agent*" OR "LLM-based multi-agent*" OR "multi-agent LLM*" OR
  "multiagent LLM*" OR "multi-agent large language model*" OR
  "tool-using LLM*" OR "tool-using agent*" OR "model context protocol" OR
  "agent-to-agent protocol" OR "Agent2Agent" OR "A2A protocol" OR
  (LLM W/3 agent*) OR ("large language model*" W/3 agent*) OR
  ("AI agent*" AND (LLM OR "large language model*")))
  AND
  (cryptograph* OR "digital signature*" OR "dual signature*" OR
  "zero-knowledge" OR "verifiable credential*" OR "verifiable provenance" OR
  "verifiable execution" OR "authenticated data structure*" OR "authenticated log*" OR
  "tamper-evident" OR "audit integrity" OR "execution integrity" OR "proof of execution" OR
  "remote attestation" OR "cryptographic attestation" OR
  "trusted execution environment*" OR "secure enclave*" OR "confidential computing" OR
  "Intel SGX" OR "Intel TDX" OR "AMD SEV*" OR "ARM TrustZone" OR
  "homomorphic encryption" OR "fully homomorphic" OR "attribute-based encryption" OR
  "proxy re-encryption" OR "functional encryption" OR "multiparty computation" OR
  "multi-party computation" OR "garbled circuit*" OR "secret sharing" OR
  "oblivious transfer" OR "Merkle tree*" OR "hash-based signature*" OR
  "post-quantum" OR "quantum-safe" OR "quantum-resilient" OR "quantum-resistant" OR
  "cryptographic agility" OR "ML-KEM" OR "ML-DSA" OR "SLH-DSA" OR "FN-DSA" OR
  "FIPS 203" OR "FIPS 204" OR "FIPS 205" OR
  "CRYSTALS-Kyber" OR "CRYSTALS-Dilithium" OR "SPHINCS+")
) AND NOT DOCTYPE(cr OR bk)
- **S2 (Authenticated Delegation, Identity & Capability Authorization — 70 raw rows; 70 unique):**
  TITLE-ABS-KEY(
  ("LLM agent*" OR "agentic AI" OR "agentic LLM*" OR "agentic workflow*" OR
  "large language model agent*" OR "LLM-based agent*" OR "LLM-powered agent*" OR
  "LLM-driven agent*" OR "multi-agent LLM*" OR "tool-using LLM*" OR
  "tool-using agent*" OR "model context protocol" OR
  "agent-to-agent protocol" OR "Agent2Agent" OR "A2A protocol" OR
  (LLM W/3 agent*) OR ("large language model*" W/3 agent*) OR
  ("AI agent*" AND (LLM OR "large language model*")))
  AND
  ("authenticated delegation" OR "secure delegation" OR

"capability-based" OR "capability token*" OR
"token binding" OR "credential binding" OR "proof of possession" OR
"SPIFFE" OR "WIMSE" OR "decentralized identifier*" OR "UMA 2.0" OR "UMA2.0" OR
((delegation OR authorization OR "access control") AND
("security protocol" OR credential* OR "policy enforcement" OR

"tool permission*" OR authentication OR "OAuth*" OR "least privilege")))

) AND NOT DOCTYPE(cr OR bk)

- **S3 (Boundary Attacks, Tool/MCP Exploits & Runtime Policy Enforcement — 103 raw rows; 103 unique):**
TITLE-ABS-KEY(
(("LLM agent*" OR "agentic AI" OR "agentic LLM*" OR "agentic workflow*" OR
"large language model agent*" OR "LLM-based agent*" OR "LLM-powered agent*" OR

"LLM-driven agent*" OR "multi-agent LLM*" OR "tool-using LLM*" OR

"tool-using agent*" OR "model context protocol" OR

(LLM W/3 agent*) OR ("large language model*" W/3 agent*) OR

("AI agent*" AND (LLM OR "large language model*")))

AND
("confused deputy" OR "tool poisoning" OR "tool hijacking" OR
"function hijacking" OR "tool squatting" OR "authorization bypass" OR

"delegation attack" OR "audit tampering" OR "tool-output injection" OR

("prompt injection" AND ("tool invocation" OR "tool calling" OR "function
↪ calling"))))

OR
(("model context protocol" OR "MCP server*" OR "MCP client*" OR
"MCP tool*" OR "MCP-enabled" OR "MCP-based" OR

(MCP AND (LLM OR "large language model*" OR "agentic AI" OR "AI agent*" OR

"tool-calling agent*" OR "tool-using agent*")) OR

"LLM agent tooling" OR "GenAI multi-agent*")

AND
("function hijacking" OR "tool squatting" OR "tool poisoning" OR
"confused deputy" OR "authorization bypass" OR "access control" OR

```
    privilege\* OR "runtime policy" OR "verifiable provenance" OR

    "audit integrity" OR "execution integrity" OR "prompt injection" OR

    "security vulnerabilit\*"))
```

) AND NOT DOCTYPE(cr OR bk)

A.1.2 IEEE Xplore Command Search (`I1a–I3b`, `Search Within: All Metadata`*).* To comply with IEEE Xplore Command Search parser limits (≤ 40 search clauses per execution, no wildcards inside quotation marks or `NEAR/3` operands, and automatic plural stemming), the three pillars were executed as five modular queries:

- **I1a (78 hits):** `("agentic AI" OR "agentic workflow" OR "LLM agent" OR "model context protocol" OR "Agent2Agent" OR "A2A protocol" OR (LLM NEAR/3 agent) OR ("large language model" NEAR/3 agent) OR ("AI agent" AND "large language model")) AND (cryptograph* OR "digital signature" OR "dual signature" OR "zero-knowledge" OR "verifiable credential" OR "verifiable provenance" OR "verifiable execution" OR "tamper-evident" OR "audit integrity" OR "execution integrity" OR "remote attestation" OR "cryptographic attestation" OR "trusted execution environment" OR "secure enclave" OR "confidential computing" OR "Intel SGX" OR "Intel TDX" OR "AMD SEV" OR "ARM TrustZone")`
- **I1b (43 hits):** `("agentic AI" OR "agentic workflow" OR "LLM agent" OR "model context protocol" OR "Agent2Agent" OR "A2A protocol" OR (LLM NEAR/3 agent) OR ("large language model" NEAR/3 agent) OR ("AI agent" AND "large language model")) AND ("homomorphic encryption" OR "fully homomorphic" OR "attribute-based encryption" OR "proxy re-encryption" OR "multiparty computation" OR "multi-party computation" OR "secret sharing" OR "Merkle tree" OR "authenticated data structure" OR "authenticated log" OR "post-quantum" OR "quantum-safe" OR "quantum-resilient" OR "quantum-resistant" OR "ML-KEM" OR "ML-DSA" OR "SLH-DSA")`
- **I2 (55 hits):** `("agentic AI" OR "agentic workflow" OR "LLM agent" OR "model context protocol" OR "Agent2Agent" OR "A2A protocol" OR (LLM NEAR/3 agent) OR ("large language model" NEAR/3 agent) OR ("AI agent" AND "large language model")) AND ("authenticated delegation" OR "secure delegation" OR "capability-based" OR "capability token" OR "token binding" OR "credential binding" OR "proof of possession" OR SPIFFE OR WIMSE OR "decentralized identifier" OR "UMA 2.0" OR ((delegation OR authorization OR "access control") AND ("security protocol" OR credential OR "policy enforcement" OR "tool permission" OR authentication OR OAuth)))`
- **I3a (22 hits):** `("agentic AI" OR "agentic workflow" OR "LLM agent" OR "tool-using agent" OR "model context protocol" OR (LLM NEAR/3 agent) OR ("large language model" NEAR/3 agent) OR ("AI agent" AND "large language model")) AND ("confused deputy" OR "tool poisoning" OR "tool hijacking" OR "function hijacking" OR "tool squatting" OR "authorization bypass" OR "delegation attack" OR "audit tampering" OR "tool-output injection" OR ("prompt injection" AND ("tool invocation" OR "tool calling" OR "function calling")))`
- **I3b (58 hits):** `("model context protocol" OR "MCP server" OR "MCP client" OR "MCP tool" OR "MCP-enabled" OR "MCP-based" OR (MCP AND (LLM OR "large language model" OR "agentic AI" OR "AI agent")) OR "LLM agent tooling" OR "GenAI multi-agent") AND ("function hijacking" OR "tool squatting" OR "tool poisoning" OR "confused deputy" OR "authorization`

```
bypass" OR "access control" OR privilege OR "runtime policy" OR "verifiable provenan
ce" OR "audit integrity" OR "execution integrity" OR "prompt injection" OR "security
vulnerability")
```

A.1.3 ACM Digital Library (ACM-S1–S3) and arXiv Atom API (ARXIV_S1a–S3). For **ACM Digital Library**, the three complete Boolean expressions supplied by the authors are recorded in `protocol/acm_arxiv_queries.md`, together with the reported export identifiers ACM-S1–S3. The author-supplied strings contain no proximity operator or explicit field prefix; the intended Title/Abstract/Keywords targeting is reported, but the original UI configuration is not independently recovered. Earlier wording asserting `"LLM agent"~3` proximity syntax and a specific field-fallback behavior is therefore withdrawn as a confirmed execution setting. The reported normalization of 2,152 raw BibTeX entries (1,465 unique) removed 608 proceedings, 9 books, 7 generated podcasts, and 728 off-topic records, yielding 113 verified ACM records; that original yield was not rerun from the newly supplied strings.

For **arXiv**, all four supplied `search_query` expressions match the constants in the recovered `arxiv_slr_supplementary_pipeline.py`, archived under `protocol/search_sources/`. The API requests use five CS category terms; the later primary/secondary-category relevance gate allows nine CS categories, as explicitly distinguished in `protocol/acm_arxiv_queries.md`. Pagination starts at `start=0`, initially requests `max_results=200`, falls back to 100 or 50, and advances by returned-entry count; no sort parameters are set. The pipeline records a 3.5 s inter-request delay and a `2026-10-15T23:59:59Z` post-retrieval publication-date filter. Compact retained exports reproduce 4,142 query returns, 3,894 base IDs, and 529 gate-passing preprints (494 new + 35 superseded); version-cutoff assessment remains a separate Stage 2 field. The live API was not rerun.

The reported publication-version cutoff and search-execution date are **15 October 2026**. Complete ACM/arXiv expressions and their provenance are now supplied in `protocol/acm_arxiv_queries.md` and machine-readable `protocol/search_sources/queries.json`. arXiv strings and request settings are corroborated by the archived code and compact occurrence exports. ACM strings are author-supplied; exact historical UI field/proximity settings and a new reproduction of the original ACM yield remain unverified.

A.2 ACM and arXiv search expressions

Release 3.6.3. Author-supplied expressions received on 3 November 2026. The four arXiv expressions match the archived pipeline constants. The three ACM expressions are supplied Boolean strings; original UI field/proximity settings remain unverified.

A.3 arXiv request and filtering configuration

Endpoint: https://export.arxiv.org/api/query. User-Agent: SLR-Agentic-Crypto-Review/1.0. Inter-request delay: 3.5 seconds. Pagination starts at 0 with max_results=200 and fallback sizes 100 and 50; the offset advances by the number of returned entries. No sortBy/sortOrder parameters are set in the archived code. Timeout: 60 seconds; up to 3 attempts per request size. The publication cutoff 2026-10-15T23:59:59Z is applied after retrieval using published_utc, rather than as an API date clause. Stage 2 separately records version-cutoff checks.

The API queries use five CS categories; the archived relevance gate allows primary/secondary categories cs.CR, cs.AI, cs.MA, cs.SE, cs.DC, cs.CL, cs.LG, cs.NI, and cs.OS. These are different steps and should not be described as identical filters. The source pipeline is archived unchanged beside queries.json; it is supplied for inspection and was not rerun against the live API.

A.4 ARXIV_S1a

Provenance: author note and matching archived pipeline. Execution fidelity: matches archived pipeline constant.

```
(cat:cs.CR OR cat:cs.AI OR cat:cs.MA OR cat:cs.SE OR cat:cs.DC) AND
```

```
(all:"LLM agent" OR all:"LLM agents" OR all:"agentic AI" OR all:"agentic" OR all:"AI agent"
↪ OR all:"AI agents" OR all:"model context protocol") AND
(all:"cryptographic" OR all:"cryptography" OR all:"cryptographically verifiable" OR
↪ all:"zero-knowledge" OR
all:"verifiable credential" OR all:"verifiable credentials" OR all:"verifiable provenance"
↪ OR all:"verifiable execution" OR
all:"tamper-evident" OR all:"remote attestation" OR all:"trusted execution environment" OR
↪ all:"confidential computing" OR all:"secure enclave")
```

A.5 ARXIV_S1b

Provenance: author note and matching archived pipeline. Execution fidelity: matches archived pipeline constant.

```
(cat:cs.CR OR cat:cs.AI OR cat:cs.MA OR cat:cs.SE OR cat:cs.DC) AND
(all:"LLM agent" OR all:"LLM agents" OR all:"agentic AI" OR all:"agentic" OR all:"AI agent"
↪ OR all:"AI agents" OR all:"model context protocol") AND
(all:"homomorphic encryption" OR all:"fully homomorphic" OR all:"attribute-based
↪ encryption" OR all:"proxy re-encryption" OR
all:"multiparty computation" OR all:"multi-party computation" OR all:"post-quantum" OR
↪ all:"quantum-safe" OR
all:"quantum-resilient" OR all:"ML-KEM" OR all:"ML-DSA" OR all:"SLH-DSA")
```

A.6 ARXIV_S2

Provenance: author note and matching archived pipeline. Execution fidelity: matches archived pipeline constant.

```
(cat:cs.CR OR cat:cs.AI OR cat:cs.MA OR cat:cs.SE OR cat:cs.DC) AND
(all:"LLM agent" OR all:"LLM agents" OR all:"agentic AI" OR all:"agentic" OR all:"AI agent"
↪ OR all:"AI agents" OR
all:"model context protocol" OR all:"Agent2Agent" OR all:"A2A protocol") AND
(all:"authenticated delegation" OR all:"secure delegation" OR all:"capability-based" OR
↪ all:"capability token" OR
all:"token binding" OR all:"credential binding" OR all:"proof of possession" OR
↪ all:"SPIFFE" OR
all:"decentralized identifier" OR all:"UMA 2.0" OR all:"security architecture" OR
↪ all:"zero-trust")
```

A.7 ARXIV_S3

Provenance: author note and matching archived pipeline. Execution fidelity: matches archived pipeline constant.

```
(cat:cs.CR OR cat:cs.AI OR cat:cs.MA OR cat:cs.SE OR cat:cs.DC) AND
(all:"LLM agent" OR all:"LLM agents" OR all:"agentic AI" OR all:"agentic" OR
↪ all:"tool-using agent" OR
all:"model context protocol" OR all:"MCP server" OR all:"MCP servers") AND
(all:"confused deputy" OR all:"tool poisoning" OR all:"tool hijacking" OR all:"function
↪ hijacking" OR
all:"tool squatting" OR all:"authorization bypass" OR all:"delegation attack" OR
↪ all:"audit tampering" OR
```

```
all:"memory exfiltration" OR (all:"model context protocol" AND (all:"prompt injection" OR
↪  all:"access control" OR all:"privilege" OR all:"runtime policy")))
```

A.8 ACM-S1

Provenance: author-supplied Boolean expression. Execution fidelity: original UI field/proximity settings not recovered.

```
("LLM agent" OR "LLM agents" OR "agentic AI" OR "agentic" OR "AI agent" OR "AI agents" OR
↪  "model context protocol")
AND
("cryptographic" OR "cryptography" OR "zero-knowledge" OR "verifiable credential" OR
↪  "verifiable provenance" OR "remote attestation" OR "trusted execution environment" OR
↪  "confidential computing" OR "secure enclave" OR "homomorphic encryption" OR
↪  "multi-party computation" OR "post-quantum")
```

A.9 ACM-S2

Provenance: author-supplied Boolean expression. Execution fidelity: original UI field/proximity settings not recovered.

```
("LLM agent" OR "LLM agents" OR "agentic AI" OR "agentic" OR "AI agent" OR "AI agents" OR
↪  "model context protocol" OR "Agent2Agent" OR "A2A protocol")
AND
("authenticated delegation" OR "secure delegation" OR "capability-based" OR "capability
↪  token" OR "token binding" OR "proof of possession" OR "SPIFFE" OR "decentralized
↪  identifier" OR "zero-trust" OR "security architecture")
```

A.10 ACM-S3

Provenance: author-supplied Boolean expression. Execution fidelity: original UI field/proximity settings not recovered.

```
("LLM agent" OR "LLM agents" OR "agentic AI" OR "agentic" OR "tool-using agent" OR "model
↪  context protocol" OR "MCP server")
AND
("confused deputy" OR "tool poisoning" OR "tool hijacking" OR "function hijacking" OR "tool
↪  squatting" OR "authorization bypass" OR "delegation attack" OR "audit tampering" OR
↪  "prompt injection" OR "runtime policy")
```

A.11 ACM field and historical-description qualification

The author note describes Title/Abstract/Keywords targeting. The supplied strings themselves do not encode field prefixes or the previously described `"LLM agent"~3` proximity operator. They are now the documented author-supplied expressions. Neither the old proximity claim nor the exact UI field configuration is established by the retained filtered export. No new ACM search was run, and the original reported yields are not asserted to have been reproduced from these strings.

A.12 Seed identity reconciliation

Use arXiv IDs to reconcile source documents whose reference numbers differ: CVA 2607.21325v3 = manuscript (Llambí-Morillas and Fernández-Fernández 2026); SAGA 2504.21034v2 = (Syros et al. 2025); MAGIQ 2605.06933v2 = (Avizheh et al. 2026); confidential-computing survey 2605.03213v2 = (Forough et al. 2026). The old (Forough

et al. 2026)/(Llambí-Morillas and Fernández-Fernández 2026)/(Syros et al. 2025) labels in the archived pipeline and author note are historical labels, not current manuscript reference numbers. The raw occurrence export identifies CVA via ARXIV_S1a, SAGA via ARXIV_S1a and ARXIV_S2, MAGIQ via ARXIV_S1a and ARXIV_S1b, and the survey via ARXIV_S1a.

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B Extraction-batch accounting

Historical numerical table retained from release 3.6.3. In the RQ3 column, Released means numeric records attached to READY claims, not eligibility for latency aggregation.

Table S24. Extraction batches and linked released/total populations.

Extraction batch	Evidence tier	Stage3_ID range	Unique studies (n = 442)	Extracted claims (n = 584)	SUPPORTED_WITH_IN_SCOPE claims	PARTIALLY / NOT_SUPPORTED claims	Released READY claims (n = 530)	Retained FLAGGED claims (n = 54)	RQ3 metrics (Released / Total)	RQ4 deps (Released / Total)
BATCH_E01	TIER_1_PEER_REVIEWED	S3-001..S3-050	50	62	57 (91.9%)	5 (8.1%)	**60 (96.8%)**	2 (3.2%)	114 / 114	13 / 13
BATCH_E02	TIER_1_PEER_REVIEWED	S3-051..S3-100	50	65	55 (84.6%)	10 (15.4%)	**58 (89.2%)**	7 (10.8%)	154 / 162	32 / 41
BATCH_E03	TIER_1_PEER_REVIEWED	S3-101..S3-150	50	60	46 (76.7%)	14 (23.3%)	**53 (88.3%)**	7 (11.7%)	153 / 187	33 / 40
BATCH_E04	TIER_1_PEER_REVIEWED	S3-151..S3-212	62	84	64 (76.2%)	20 (23.8%)	**72 (85.7%)**	12 (14.3%)	247 / 270	42 / 56
BATCH_E05 *(Pilot)*	TIER_2_ARXIV_PREPRINT	S3-213..S3-236	24	39	34 (87.2%)	5 (12.8%)	**36 (92.3%)**	3 (7.7%)	91 / 96	19 / 21
BATCH_E06	TIER_2_ARXIV_PREPRINT	S3-237..S3-286	50	68	61 (89.7%)	7 (10.3%)	**67 (98.5%)**	1 (1.5%)	66 / 70	26 / 26
BATCH_E07	TIER_2_ARXIV_PREPRINT	S3-287..S3-336	50	67	52 (77.6%)	15 (22.4%)	**57 (85.1%)**	10 (14.9%)	54 / 67	24 / 28
BATCH_E08	TIER_2_ARXIV_PREPRINT	S3-337..S3-386	50	65	53 (81.5%)	12 (18.5%)	**60 (92.3%)**	5 (7.7%)	59 / 65	28 / 31
BATCH_E09	TIER_2_ARXIV_PREPRINT	S3-387..S3-442	56	74	60 (81.1%)	14 (18.9%)	**67 (90.5%)**	7 (9.5%)	70 / 80	24 / 27

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Table S24. Extraction batches and linked released/total populations. (continued)

Extraction batch	Evidence tier	Stage3_ID range	Unique studies (n = 442)	Extracted claims (n = 584)	SUPPORTED_WITHIN_SCOPE claims	PARTIALLY / NOT_SUPPORTED claims	Released READY claims (n = 530)	Retained FLAGGED claims (n = 54)	RQ3 metrics (Released / Total)	RQ4 deps (Released / Total)
Total (All Batches)	**Tier 1 (212) + Tier 2 (230)**	**S3-001..S3-442**	**442**	**584**	**482 (82.5%)**	**102 (17.5%)**	**530 (90.8%)**	**54 (9.2%)**	**1,008 / 1,111 (90.7%)**	**241 / 283 (85.2%)**

C Focused formal-evidence appraisals

These are scoped records transcribed from the reviewed extraction, not independently replayed tool results. Category labels do not certify implementation correspondence or proof replication.

Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping.

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0001 (SLR-M-0219)	**CapSeal** (Jin et al., 2026) (Jin et al. 2026)	SECURITY_REDUCTION	FHE IND-CPA (TFHE-rs) + conditional PRF/MAC capability binding	B1;B2;B3;B4	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Operator blindness proved under IND-CPA, but prototype benchmark uses a 16-round non-cryptographic mult-XOR stand-in (~9–10 s/request) while p. 15 asserts 256-bit security; LowMC requires ~2.6 h
CLM-0010 (ARXIV-S-0407)	**BAM** (Guo et al. 2026)	SECURITY_REDUCTION	Source-described PRF-based steganographic reduction	B2;B3;B4	CONTROLLED_BENCHMARK, SUPPORTED_WITHIN_SCOPE	The guarantee assumes secret seed/nonce synchronization and sender sampling control. Zero errors in 1,000 trials is an observed result, not zero population error; finite-precision leakage and altered channels remain outside the ideal guarantee.
CLM-0028 (ACM-S-0095)	**Agentic IAM** (Zouari, 2025) (Zouari 2025)	TOOL_SUPPORTED_VERIFICATION	Bayesian belief update + MDP CVaR risk-constrained optimization	B2;B3	PROOF_OF_CONCEPT (PARTIALLY_SUPPORTED)	Assumes rational non-colluding adversary and trusted state-compromise detector; Table 2 single-VM action reduction (46 → 24) conflicts with prose (39%–46%)
CLM-0052 (ARXIV-S-0010)	**TMA-NM** (Louck, 2026) (Louck 2026)	TOOL_SUPPORTED_VERIFICATION	TLA+ / TLC model checking (3,270 reachable states) of non-malleable origin labels	B2;B3	CONTROLLED_BENCHMARK (SUPPORTED_WITHIN_SCOPE)	Machine-checked guarantee (0/4,032 consequential memory-poisoning attacks) requires trusted channel-origin labeling (A1) and independent corroborators

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Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping. (continued)

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0060, CLM-0061 (ARXIV-S-0019)	**PunkGo** (Zhang, 2026) (Zhang 2026)	LOGICAL_AXIOMS_WITH_PROOF	RFC 6962 SHA-256 Merkle tree invariants + energy/hold state-machine axioms	B4 (0060), B3 (0061)	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Proves append-only history integrity (1.26 ms median execute) under a single trusted Rust kernel committer; excludes root-level SQLite rewrite and external tool correctness
CLM-0079 (ARXIV-S-0044)	**Omega** (Bodea et al., 2025) (Bodea et al. 2025)	TOOL_SUPPORTED_VERIFICATION	**Tamarin** symbolic multiset-rewriting proof of differential CVM/CGPU attestation	B3	RESEARCH_PROTOTYPE_SIMULATOR (SUPPORTED_WITHIN_SCOPE)	Symbolic secrecy/correspondence holds for attested channels (39.8 ms policy check), but prototype evaluation disables VMPL isolation and trusted boot (p. 12)
CLM-0095 (ACM-S-0019)	**CapScope** (Bouras et al., 2026) (Bouras et al. 2026)	LOGICAL_AXIOMS_WITH_PROOF	Operational semantics & monotonic attenuation axioms (Mint, Derive, Invoke)	B1;B2;B3	CONTROLLED_BENCHMARK (SUPPORTED_WITHIN_SCOPE)	Eliminates ambient authority in coding agents, but relies on a trusted host policy compiler and increases mean wall-clock task time from 194 s to 316 s
CLM-0109 (ACM-S-0047)	**AgentOS VTRM** (Zhou & Jia, 2026) (J. Zhou and Jia 2026)	TOOL_SUPPORTED_VERIFICATION	**TLA+** formal specification of typed tool mediation and fail-closed world-effect gating	B3	RESEARCH_PROTOTYPE_SIMULATOR (SUPPORTED_WITHIN_SCOPE)	Proves abstract fail-closed mediation invariants, conditional on accurate security-label derivation and complete host-side tool wrapping
CLM-0197 (SLR-M-0104)	**BT-PKEET** (Z. Li et al. 2026)	SECURITY_REDUCTION	Source-described pairing-based OW-CCA / IND-CCA reductions	B2;B3;B4	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	The reduction and primitive benchmarks do not establish full-workflow assurance. The reported Type-A subgroup order is 128 bits; this is not evidence of 128-bit security. Repeated-query leakage and token pooling remain outside the assessed scope.

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Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping. (continued)

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0280 (SLR-M-0262)	**SecuAudit** (Shi et al., 2026) (Y. Shi et al. 2026)	SECURITY_REDUCTION	Game-based auditing soundness under CDH/DL and (t, m) Shamir threshold honesty	B2;B3;B4	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Uses 160-bit subgroup ($\approx$ 80-bit security); game soundness does not cover adaptive TPA privacy or on-chain approval-count authentication
CLM-0320 (SLR-M-0327)	**VF-RABPRE** (Cai et al., 2026) (Cai et al. 2026)	SECURITY_REDUCTION	IND-CPA under q-set consistent BDHE + DLEQ NIZK and Groth16/PLONK fairness	B2;B3	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Proves CPA confidentiality (Groth16 prove 1.49–1.97 s, verify 1–2 ms), not IND-CCA or atomic payment settlement; pairing curves are pre-quantum
CLM-0372 (ARXIV-S-0109)	**Source-Channel HMAC** (Leong 2026)	SECURITY_REDUCTION (recorded)	External reference monitor and HMAC-authenticated capability checks	B2;B3	CONTROLLED_BENCHMARK, SUPPORTED_WITHIN_SCOPE	Tested unauthorized execution is blocked under the monitor's assumptions. Dynamic capability acquisition remains an operational limit: the recorded appraisal reports legitimate throughput falling to 64% on weaker models.
CLM-0383 (ARXIV-S-0125)	**APB(k)** (Fernandez, 2026) (Fernandez 2026)	SECURITY_REDUCTION	Ed25519 EUF-CMA k-of-n multi-principal signature binding (RFC 8785 JCS) + HMAC log	B1;B2;B4	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Achieves 100% tamper/forgery detection across 7,000 multisigner events and 200 attack trials; relies on uncompromised human signer keys and caller uniqueness store
CLM-0395 (ARXIV-S-0141)	**Agentic Witnessing** (Rowstron 2026)	TOOL_SUPPORTED_VERIFICATION (recorded)	Corpus commitments, restricted-verdict interface, and source-described invariants	B3;B4	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	Recorded evaluation uses Docker without SEV; TEE/attestation is emulated. The leakage bound concerns the verdict channel under a session budget, excluding timing, metadata and accumulated sessions. Remote inference and semantic judgment remain trusted.

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Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping. (continued)

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0401 (ARXIV-S-0160)	**SAGA** (Syros et al. 2025)	TOOL_SUPPORTED_VERIFICATION	Source-described ProVerif secrecy/authentication model	B1;B2;B3	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	Policy-authorized contacts remain conditional on trusted providers, registration and key custody. Valid-token malicious communication is admitted within quota/expiry. The appraisal notes inconsistent Curve25519/X25519 signature terminology; no exact signature suite is inferred from that wording.
CLM-0420, CLM-0422 (ARXIV-S-0203)	**AITH** (Z. Chen 2026)	TOOL_SUPPORTED_VERIFICATION (recorded; both FLAGGED)	Source asserts Tamarin lemmas; model/log artifacts were not inspected in the recorded appraisal	B1;B2;B3;B4	RESEARCH_PROTOTYPE_SIMULATOR; partial/support status differs by claim	The benchmark substitutes HMAC-SHA256 for intended ML-DSA. Its timings cannot support lattice-signature performance. Reported verification and red-team activity do not remove this implementation mismatch.
CLM-0430 (ARXIV-S-0222)	**Z^2-ACT** (Khowaja et al., 2026) (Khowaja et al. 2026)	SECURITY_REDUCTION	Analytical game bound on zero-trust intent contracts and near-RT RAN scope locks	B1;B2;B3	RESEARCH_PROTOTYPE_SIMULATOR (SUPPORTED_WITHIN_SCOPE)	Evaluates 6G Open RAN multi-agent control in simulation; relies on synchronized scope-lock clocks and trusted near-RT RIC gateway
CLM-0436 (ARXIV-S-0227)	**A402** (Li et al., 2026) (Y. Li, L. Wang, et al. 2026)	SECURITY_REDUCTION	Adaptor-signature atomicity reduction + TEE execution attestation for agentic commerce	B2;B3;B4	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Binds payment release to attested service execution (61.24 ms off-chain vs. 2–120 s on-chain); depends on TEE key sealing and dispute-window liveness
CLM-0452 (ARXIV-S-0254)	**Sentinel-Agent** (Patil 2026)	TOOL_SUPPORTED_VERIFICATION	Source-reported finite-state TLA+ checks	B1;B2;B3;B4	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	Probabilistic intent property P2 is outside deterministic checking. The appraisal records symmetric-key custody and simulated threshold signing; it does not validate an implemented Ed25519 multi-principal chain.

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Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping. (continued)

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0456 (ARXIV-S-0259)	**SUDP** (Yu et al., 2026) (X. Yu et al. 2026)	SECURITY_REDUCTION	Game-based secret-use delegation reduction (WebAuthn/FIDO2 challenge + custodian broker)	B1;B2;B3;B4	FORMAL_MODEL_ONLY (SUPPORTED_WITHIN_SCOPE)	Proves untrusted agent cannot extract raw secrets under honest custodian and hardware authenticator display; no empirical runtime benchmark reported
CLM-0495 (ARXIV-S-0328)	**A2ABreak** (Lotfi et al. 2026)	TOOL_SUPPORTED_VERIFICATION	Alloy for context ownership and skill attestation; TLA+/TLC for delegation provenance (source Section VI)	B2;B3;B4	FORMAL_MODEL_ONLY, SUPPORTED_WITHIN_SCOPE	Three modeled findings/mitigations support specification-level counterexamples and restored model invariants. The source explicitly excludes ProVerif. R3 corrects the historical tool attribution; no model replay or live SDK exploitation is established here.
CLM-0498 (ARXIV-S-0330)	**enclawed** (Metere 2026)	TOOL_SUPPORTED_VERIFICATION (recorded)	Source-described static/refinement checks and bounded skill-evidence search	B1;B2;B3;B4	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	Companion formal theorems were not reproduced. JavaScript bootstrap, parser coverage, entry-point mediation and operator-supplied OS controls remain trusted; no ProVerif implementation proof is established by the supplied appraisal.
CLM-0535 (ARXIV-S-0414)	**Opal** (Kaviani et al., 2026) (Kaviani et al. 2026)	SECURITY_REDUCTION	Obliviousness & IND-CPA composition of Intel TDX enclave + ORAM private memory	B2;B3	DEPLOYMENT_LIKE_EVALUATION (SUPPORTED_WITHIN_SCOPE)	Proves memory-access-pattern obliviousness (2.32 s vs. 1.48 s plaintext); relies on TDX hardware TCB and fixed oblivious access schedule
CLM-0542 (ARXIV-S-0427)	**PCC-LLM** (Koomullil, 2026) (Koomullil 2026)	TOOL_SUPPORTED_VERIFICATION	**Lean 4** proof-carrying structured certificates (22 certificate types, zero sorry axioms)	B2;B3;B4	PROOF_OF_CONCEPT (PARTIALLY_SUPPORTED)	Pilot C blocks 12/18 adversarial proposals; five allowed probes satisfy the deletion budget and one concerns an omitted mtime predicate. The source retains HC1 failure under its registered denominator. These are not six violations of the formal checker specification. Compiled certificate claims remain source-reported, not replayed here.

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Table S25. Scoped appraisals of the 29 claims in the focused formal-evidence grouping. (continued)

Claim_ID (Source_Record_ID)	Study & citation	Formal evidence type	Verification tool or cryptographic hardness assumption	Target boundaries (Boundary_Codes)	Implementation maturity & support status	Critical TCB boundary or semantic-formal gap identified in Stage 3 audit
CLM-0547 (ARXIV-S-0432)	**ConLeash** (Li et al., 2026) (Y. Li, Y. Chen, et al. 2026)	TOOL_SUPPORTED_VERIFICATION	**Datalog** decidable consent-lattice refinement and least-privilege policy engine	B1;B2;B3	PROOF_OF_CONCEPT (PARTIALLY_SUPPORTED)	The source reports 8.2 ms Datalog evaluation and 4.69 s LLM predicate extraction. It also describes type checks, taint propagation and generated policy tests. These controls matter, but do not by themselves establish faithful representation of every natural-language constraint.
CLM-0551 (ARXIV-S-0440)	**A-VIP / AP2** (Louck et al., 2026) (Louck et al. 2026)	TOOL_SUPPORTED_VERIFICATION	**TLA+ / TLC** model checking of session-, cart-, merchant-, and nonce-bound payment credentials	B1;B2;B3	PROOF_OF_CONCEPT (SUPPORTED_WITHIN_SCOPE)	Eliminates Cart/Merchant/Vault Whisper attacks (56%–90% ASR on baseline AP2), but SelectionWhisper within user budget limits still requires human display confirmation
CLM-0565 (ARXIV-S-0464)	**AEGIS-Router** (Xie et al. 2026)	TOOL_SUPPORTED_VERIFICATION	Scoped ProVerif model, source-described reductions and Nitro implementation	B2;B3;B4	PROOF_OF_CONCEPT, SUPPORTED_WITHIN_SCOPE	The 5.7 ms median comes from a local benchmark image distinct from the real-provider security image. Hardware roots, audited code and faithful relay remain trusted; full-program noninterference is not established.

D PQC migration appraisals

The entries preserve the source-specific scope and QA qualifications. Primitive implementation, agent-workload evaluation, and operational deployment are distinct stages.

Table S26. Recorded appraisals of the 24 claims at PQC migration Stages 1--5.

Stage & Claim_IDs	Source_Record_ID (Tier)	Study & system name	Cryptographic / PQC construction evaluated or proposed	Reported performance overhead & key/signature artifact metrics	Stage 3 audit finding & residual PQC gap (Claim_Support_Status)
STAGE_5 (CLM-0335, CLM-0336)	ARXIV-S-0057 (Tier 2)	**MAGIQ** (Avizheh et al., 2026) (Avizheh et al. 2026)	XMSS root authority + hybrid X25519/ML-KEM-768 PQ-mTLS + ML-DSA-65 governance signatures + **amortized HMAC-SHA256 session policy tickets**	Real PQ-mTLS handshake + ML-DSA-65 signing measured on multi-agent governance workloads; session ticket amortization eliminates per-call lattice signature bottleneck	**SUPPORTED_WITHIN_SCOPE**: Most complete PQC agent governance architecture in the corpus; demonstrates that session-ticket amortization resolves ML-DSA per-call bandwidth/CPU overhead
STAGE_5 (CLM-0392, CLM-0393, CLM-0394)	ARXIV-S-0140 (Tier 2)	**QSC** (Bishwas et al., 2026) (Bishwas et al. 2026)	Sidecar PQC-TLS 1.3 (ML-KEM-768 / Kyber768 + ML-DSA-65 / Dilithium3 + AES-256-GCM) + source-claimed QRNG throughput (device/rate discrepancy in the appraisal) + modeled BB84 QKD	**Microbenchmark (Table 2):** ML-KEM encap 0.082 ms (vs. 0.018 ms classical), decap 0.014 ms, keygen 0.11 ms; ML-DSA sign 0.34 ms (vs. 0.01 ms), verify 0.08 ms (vs. 0.03 ms); KEM PK 1,184 B, CT 1,088 B; Sig PK 1,952 B (vs. 32 B), **Sig 3,309 B (vs. 64 B, +5,070%)**. The reported 297 ms cloud baseline is a configuration-specific comparator	**CLM-0392 and CLM-0394: SUPPORTED_WITHIN_SCOPE, READY; CLM-0393: PARTIALLY_SUPPORTED, FLAGGED.** The retained sub-scopes report lattice primitive timings; the appraisal notes inconsistent printed overhead percentages and QRNG device throughput. They also report the 51.7x signature-size expansion (3,309 B vs. 64 B); the QKD layer is simulated (10 ms), not physical quantum fiber. The grouped study row does not release the flagged sibling claim
STAGE_4 (CLM-0021)	SLR-M-0244 (Tier 1)	**Quantum-Enhanced Cloud** (Sudarsan et al., 2026) (Sudarsan et al. 2026)	liboqs (OQS) Kyber / Dilithium integration + Amazon Braket BB84 / Grover simulation	Reports 26.9 TPS vs. 16.4 TPS baseline, mean latency 248 ms vs. 427 ms (p95 = 396 ms vs. 683 ms)	**PARTIALLY_SUPPORTED (AD;UG)**: OQS primitive integration is accompanied by an unisolated 89% vulnerability-reduction claim and simulated quantum speedup without reporting exact NIST parameter sets

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Table S26. Recorded appraisals of the 24 claims at PQC migration Stages 1--5. (continued)

Stage & Claim_IDs	Source_Record_ID (Tier)	Study & system name	Cryptographic / PQC construction evaluated or proposed	Reported performance overhead & key/signature artifact metrics	Stage 3 audit finding & residual PQC gap (Claim_Support_Status)
STAGE_4 (CLM-0207)	SLR-M-0127 (Tier 1)	**CAESF-PQC-APKAI** (Palla, 2026) (Palla 2026)	Hybrid RSA/ECC + Kyber/Dilithium automated PKI and blockchain certificate logging	Table IX reports total 32.6 ms (Kyber keygen 2.3 ms, encap 1.8 ms, decap 2.0 ms; Dilithium sign 4.6 ms, verify 2.7 ms) in a simulated enterprise environment	**PARTIALLY_SUPPORTED (AD;UG)**: Reports 99.3% *"quantum attack resistance"* without an operational cryptographic definition, library version, or confidence intervals
STAGE_3 (CLM-0420, CLM-0421, CLM-0422)	ARXIV-S-0203 (Tier 2)	**AITH** (Chen, 2026) (Z. Chen 2026)	Continuous human-AI delegation protocol claiming ML-DSA-87 post-quantum signatures	Reports 0.21 μs signature verification and 5.4 μs 6-check engine latency	**PARTIALLY_SUPPORTED (CLM-0420/0421)**: Full-text code audit reveals ML-DSA-87 was **simulated using HMAC-SHA256** (0.21 μs is symmetric hashing, not lattice verification)
STAGE_3 (CLM-0271)	SLR-M-0245 (Tier 1)	**Quantum-Resilient FinTech** (Yerra & Cherukapalle, 2026) (Yerra and Cherukapalle 2026)	Monte Carlo workflow simulation with abstract PQC/zero-trust parameters	Simulated ASR = 6.9% vs. 38.6% baseline; 99.6% verified actions	**PARTIALLY_SUPPORTED (AD;UG)**: Quantum-resilience conclusion is an input assumption of the simulation rather than an implemented cryptographic evaluation
STAGE_2 (CLM-0264)	SLR-M-0230 (Tier 1)	**ContextGuard-MCP-PQC** (Gentyala et al., 2026) (Gentyala et al. 2026)	Analytical proposal for ML-KEM + X25519 and ML-DSA + ECDSA dual-PKI governance in MCP	Cites external 2,272 B handshake estimate from literature; NR for own measurements	**PARTIALLY_SUPPORTED (AD;UG)**: Purely analytical proposal; contains no prototype implementation or empirical MCP benchmark
STAGE_2 (CLM-0011)	SLR-M-0317 (Tier 1)	**TAEA / TOAEM** (Jingar, 2026) (Jingar 2026)	Conceptual trust-orchestration score combining cloud assurance and PQC predicates	Claims 90%–95% trust accuracy and 96% quantum-attack detection without raw tables	**PARTIALLY_SUPPORTED (AD;UG)**: Conceptual architecture with undefined quantum-detection metrics and no instantiated PQC scheme

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Table S26. Recorded appraisals of the 24 claims at PQC migration Stages 1--5. (continued)

Stage & Claim_IDs	Source_Record_ID (Tier)	Study & system name	Cryptographic / PQC construction evaluated or proposed	Reported performance overhead & key/signature artifact metrics	Stage 3 audit finding & residual PQC gap (Claim_Support_Status)
STAGE_1 (CLM-0030, CLM-0132)	ACM-S-0102	**TrustOrch** (Hu et al. 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0030: SUPPORTED_WITHIN_SCOPE / READY; CLM-0132: SUPPORTED_WITHIN_SCOPE / READY
STAGE_1 (CLM-0170)	SLR-M-0058	**Agentic CDN** (Kolla et al. 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0170: SUPPORTED_WITHIN_SCOPE / READY
STAGE_1 (CLM-0171, CLM-0172)	SLR-M-0059	**Agentic Cybersecurity** (Gonzalez-Gorrin and Prieto-Blazquez 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0171: SUPPORTED_WITHIN_SCOPE / READY; CLM-0172: SUPPORTED_WITHIN_SCOPE / READY
STAGE_1 (CLM-0291, CLM-0292)	SLR-M-0272	**SPI-Net** (Alhwikem et al. 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0291: SUPPORTED_WITHIN_SCOPE / READY; CLM-0292: SUPPORTED_WITHIN_SCOPE / READY
STAGE_1 (CLM-0047)	ARXIV-S-0003	**NovaFabric** (Ardebili 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0047: SUPPORTED_WITHIN_SCOPE / READY

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Table S26. Recorded appraisals of the 24 claims at PQC migration Stages 1--5. (continued)

Stage & Claim_IDs	Source_Record_ID (Tier)	Study & system name	Cryptographic / PQC construction evaluated or proposed	Reported performance overhead & key/signature artifact metrics	Stage 3 audit finding & residual PQC gap (Claim_Support_Status)
STAGE_1 (CLM-0054)	ARXIV-S-0013	**HBHC** (Deochake 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0054: SUPPORTED_WITHIN_SCOPE / READY
STAGE_1 (CLM-0039)	ARXIV-S-0229	**AgentFacts** (Grogan 2025)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0039: PARTIALLY_SUPPORTED / FLAGGED
STAGE_1 (CLM-0014)	ARXIV-S-0361	**Social Contract** (Ruan 2026)	Classical mechanism with a proposed future PQC migration; see claim-specific mechanism and scope in data/claims.csv	No implemented PQC benchmark assigned at Stage 1	CLM-0014: PARTIALLY_SUPPORTED / FLAGGED

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E Recorded claim-release dispositions

The table records the historical 107-candidate appraisal, not a new source review. READY is a claim-level disposition; attached metrics retain their own eligibility and conflict status. The underlying candidate matrix is not distributed. Detailed examples remain in the accompanying supplementary package.

Table S27. Historical release dispositions for the 107 candidate claims.

Decision zone & forensic subcategory (reported decision-zone taxonomy; internal candidate matrix not distributed)	Total claims (n = 107)	Tier 1 (n = 53)	Tier 2 (n = 54)	Linked RQ3 metrics (n = 179)	Linked RQ4 deps (n = 72)	Operational release disposition & methodological justification
P1 (Green Zone — Priority 1): SUPPORTED_WITHIN_SCOPE with confirmed online-first 2026 publication (SLR-M-0139 CLM-0217; SLR-M-0289 CLM-0301–0303)	**4** (3.7%)	4	0	10	3	**Promoted to READY (VERIFIED):** DOI (10.1016/j.eswa.2026.133781) and Stage 2 Scopus indexing (< 15 Oct 2026) confirm online-first 2026 publication prior to the cutoff despite 2027 volume cover dates
P2 (Yellow-A Zone — Priority 2): PARTIALLY_SUPPORTED with real prototype/benchmark/formal proof and an explicitly fenced usable claim scope; linked metrics retain their separate dispositions	**49** (45.8%)	21	28	78	27	**Promoted to READY (ADJUDICATED):** Strictly fenced in Assessed_Claim_Scope to the empirically/formally verified sub-mechanism (25 PoC, 14 CONTROLLED_BENCHMARK, 7 SIMULATOR, 2 DEPLOYMENT_EVAL, 1 THEORETIC_PROOF)
P3 (Yellow-B Zone — Conditional QA Exclusion): PARTIALLY_SUPPORTED with real prototype/simulator but unresolved RQ3 metric conflict or unclear workload denominator	**11** (10.3%)	9	2	38	13	**Retained at FLAGGED (EXTRACTED):** Held outside the primary release because of source-level table-vs-prose discrepancies (CLM-0233, CLM-0250, CLM-0282, CLM-0314, CLM-0492) or simulation denominators (CLM-0021, CLM-0057, CLM-0156, CLM-0159, CLM-0207, CLM-0271) or unresolved methods in the recorded appraisal
R1 (Red Zone — Hard Reject): Broken cryptographic signature construction (NOT_SUPPORTED CLM-0446 + sibling CLM-0445 on ARXIV-S-0245)	**2** (1.9%)	0	2	2	1	**Permanently Rejected (FLAGGED):** Equation 3 defines a digital signature as an unkeyed SHA-256 digest; hard-blocked by stage3_quality.py

Continued on next page

Table S27. Historical release dispositions for the 107 candidate claims. (continued)

Decision zone & forensic subcategory (reported decision-zone taxonomy; internal candidate matrix not distributed)	Total claims (n = 107)	Tier 1 (n = 53)	Tier 2 (n = 54)	Linked RQ3 metrics (n = 179)	Linked RQ4 deps (n = 72)	Operational release disposition & methodological justification
R2 (Red Zone — Hard Reject): Simulated cryptographic/formal primitives (ML-DSA via HMAC, synthetic Z3 log, FHE/MPC stubs, weak PRNG) or [specify] table placeholders	**7** (6.5%)	1	6	6	5	**Permanently Rejected (FLAGGED):** Prevents simulated PQC/formal timings (CLM-0420–0422, CLM-0041, CLM-0042, CLM-0147, CLM-0474) from contaminating empirical quartile syntheses
R3 (Red Zone — Hard Reject): Direct internal table-vs-prose/formula contradiction, train-on-test leakage, unit error (ms vs. ns), or *illustrative* non-empirical table	**12** (11.2%)	6	6	29	6	**Permanently Rejected (FLAGGED):** Source self-contradictions (CLM-0577, CLM-0575, CLM-0418, CLM-0230, CLM-0235, CLM-0267, CLM-0268, CLM-0062, CLM-0196, CLM-0184, CLM-0393, CLM-0500) cannot be repaired without altering source claims
R4 (Red Zone — Hard Reject): Unimplemented CONCEPTUAL_ARCHITECTURE (19 claims, incl. uncompiled Lean 4 §10 card CLM-0544) or unquantified 1-case demo (3 claims) that overclaim	**22** (20.6%)	12	10	16	17	**Permanently Rejected (FLAGGED):** Preserved as evidence of the design-to-implementation gap in RQ2, but excluded from verified empirical release
Total Audited FLAGGED Candidates	**107 (100.0%)**	**53**	**54**	**179**	**72**	**53 Promoted to READY (P1 + P2 → 530 READY total) vs. 54 Retained at FLAGGED (43 Red Zone R1–R4 + 11 P3)**

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